

The antigenic landscape of human influenza N2 neuraminidases from 2009 until 2017

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Abstract

Human H3N2 influenza viruses are subject to rapid antigenic evolution which translates into frequent updates of the composition of seasonal influenza vaccines. Despite these updates, the effectiveness of influenza vaccines against H3N2-associated disease is suboptimal. Seasonal influenza vaccines primarily induce hemagglutinin-specific antibody responses. However, antibodies directed against influenza neuraminidase (NA) also contribute to protection. Here, we analyzed the antigenic diversity of a panel of N2 NAs derived from human H3N2 viruses that circulated between 2009 and 2017. The antigenic breadth of these NAs was determined based on the NA inhibition (NAI) of a broad panel of ferret and mouse immune sera that were raised by infection and recombinant N2 NA immunization. This assessment allowed us to distinguish at least 4 antigenic groups in the N2 NAs derived from human H3N2 viruses that circulated between 2009 and 2017. Computational analysis further revealed that the amino acid residues in N2 NA that have a major impact on susceptibility to NAI by immune sera are in proximity of the catalytic site. Finally, a machine learning method was developed that allowed to accurately predict the impact of mutations that are present in our N2 NA panel on NAI. These findings have important implications for the renewed interest to develop improved influenza vaccines based on the inclusion of a protective NA antigen formulation.

eLife assessment

This study presents **valuable** data on the antigenic properties of neuraminidase proteins of human A/H3N2 influenza viruses sampled between 2009 and 2017. The antigenic properties are found to be generally concordant with genetic groups. However, the evidence for claims concerning the molecular basis of the antigenic differences remains **incomplete**, as the computational methods have been insufficiently validated.

Introduction

Neuraminidase (NA) and hemagglutinin (HA) constitute the two major antigenic proteins at the surface of influenza A and B virions (Uyeki et al., 2022 [↗](#)). While HA mediates binding to sialic acids and membrane fusion, NA catalyses the removal of sialic acid residues from sialylated glycoproteins and glycolipids. This complementary role of HA and NA enables influenza virions to penetrate the mucus barrier that lines the respiratory epithelium and newly formed virions to be released from the infected cell (de Vries et al., 2020 [↗](#); Matrosovich et al., 2004 [↗](#); Palese et al., 1974 [↗](#)). Immunity against influenza is associated with the presence of anti-HA antibodies in serum, as determined by a hemagglutination inhibition assay. In line with this, the composition of licensed influenza vaccines is primarily focused on the antigenic characteristics of HA as the protective vaccine antigen (Cox, 2013 [↗](#)). Multiple studies, however, have underscored the protective potential of anti-NA immune responses as determined by NA inhibition (NAI) assays (Couch et al., 2013 [↗](#); Maier et al., 2019 [↗](#); Memoli et al., 2016 [↗](#); Monto and Kendal, 1973 [↗](#); Monto et al., 2015 [↗](#); Ng et al., 2019 [↗](#); Weiss et al., 2019 [↗](#)). There is also evidence that pre-existing anti-N2 immunity induced by prior exposure to H2N2 viruses that circulated in the population from 1957 to 1968, tempered the impact of the H3N2 pandemic virus that emerged in 1968 (Eickhoff and Meiklejohn, 1969 [↗](#); Jester et al., 2020 [↗](#); Monto and Kendal, 1973 [↗](#); Viboud et al., 2005 [↗](#)). Further, the pharmacological inhibition of NA activity is a well-established therapeutic strategy to try to reduce influenza severity and duration (Leneva et al., 2000 [↗](#); Yen et al., 2005 [↗](#)). In addition, antibodies with NAI activity can protect against influenza virus infection in pre-clinical models (Chen et al., 2018 [↗](#); Deroo et al., 1996 [↗](#); Rosu et al., 2022 [↗](#); Wan et al., 2018 [↗](#); Wohlbald et al., 2017 [↗](#)).

Since its appearance in 1968, human H3N2 viruses have been evolving faster than the co-circulating H1N1 and influenza B viruses (Bedford et al., 2014 [↗](#); Potter et al., 2019 [↗](#)). The overall Influenza virus vaccine effectiveness varies between seasons and, despite frequent updates of the H3N2 HA component, the effectiveness of influenza vaccines against H3N2-associated disease ranges between 5 and 39%, which is lower than the effectiveness against H1N1 and influenza B (Belongia et al., 2016 [↗](#)). In this context the use of NA as a complementary vaccine antigen might improve influenza vaccine effectiveness, for example by compensating for HA-based antigenic mismatches in circulating influenza virus strains (Krammer et al., 2018 [↗](#)).

Phylogenetic analyses of HA and NA genes of human H3N2 viruses revealed the asynchronous evolution of the 2 proteins. In addition, the evolutionary rate as determined by the accumulation of mutations over time, is estimated to be 1.5 times lower for NA than for HA (Westgeest et al., 2012 [↗](#)). Despite the strong evidence that immune responses directed against NA can contribute to protection against influenza, a comprehensive insight in the antigenic breadth and evolution of the H3N2 NA antigen is lacking (Nachbagauer et al., 2017 [↗](#); Rajendran et al., 2017 [↗](#); Smith et al., 2004 [↗](#); Westgeest et al., 2012 [↗](#)).

Here, we determined the antigenic landscape of N2 NAs derived from human H3N2 strains that were isolated between 2009 and 2017. This analysis allowed the identification of at least 4 N2 NA antigenic groups and to estimate the weight of amino acid changes on their contribution to NA antigenic drift.

Generation of H3N2-derived NA immune sera in ferrets and mice

Our aim was to determine the antigenic relationship between NAs derived from human H3N2 viruses that circulated during nearly one decade. We first selected 44 H3N2 neuraminidase protein sequences obtained via GISAID, derived from viruses that were isolated between 2009 and 2017 (Elbe and Buckland-Merrett, 2017 [↗](#)). Hierarchical clustering with minimax linkage was employed to ensure the set was representative of overall protein sequence diversity (Bien and Tibshirani, 2011 [↗](#)). Next, 44 recombinant H1N2 viruses were generated that carried the internal and HA genes from PR8 virus and the NA genes of the selected H3N2 viruses. In addition, the corresponding tetrabrachion-stabilized recombinant N2 NA proteins were produced in Chinese hamster ovary (CHO) cells and purified, and are referred to as NACHOs (Catani et al., 2022 [↗](#)). All purified NACHOs were tetrameric based on SEC-MALS analysis. The enzymatic activity of the NACHOs was determined based on the hydrolysis of the fluorogenic small molecule substrate 4-methylumbelliferyl-N-acetylneuraminic acid (MUNANA). The NACHOs were active with a specific activity ranging from 0.27 to 27.48 nmole/min/ μ g, indicating correct folding and assembly of the NA head domains (Table S1 [↗](#)).

Immune sera against all 44 H3N2-derived NAs were generated in pairs of ferrets by a primary infection with the respective H1N2 reassortant viruses. This prime by infection was used to avoid the use of an adjuvant upon subsequent boost with the corresponding NACHOs. On day 28 after primary infection, all animals were boosted by intramuscular injection of homologous NACHO, except for the ferrets that had been primed by infection with A/Stockholm/15/2014 and A/Estonia/91621/2015, which were re-inoculated with the same respective H1N2 reassortant viruses. These two pairs of ferrets were not boosted with homologous NACHO because the corresponding NACHOs were not available at the time of the experiment. Blood samples were taken from the ferrets on days 14 and 42 after (primary) infection (Figure 1a [↗](#)). Priming by H1N2 infection resulted in seroconversion against the respective immobilized homologous NACHOs (Figure 1b [↗](#)). Subsequent immunization with recombinant NACHOs boosted the NACHO-specific IgG titers in day 42 serum samples. In contrast, serum samples from the ferrets that had been re-inoculated with H1N2 reassortant carrying NA derived from A/Stockholm/15/2014 or A/Estonia/91621/2015, showed no further increase in titers, presumably because prior exposure with the same H1N2 virus had induced a sterilizing immune response (Figure 1b [↗](#)).

We also generated NA immune sera in BALB/c mice. Groups of 5 mice were immunized two times, with a two-week interval, by intramuscular injection with 1 μ g of the respective 43 NACHOs adjuvanted with AF03 (the NACHO A/West-Virginia/17/2012 was not available for mice immunization and ELISAs). Blood was collected from the immunized mice and serum prepared on day 28 after the prime immunization (Figure 1c [↗](#)). NACHO-specific IgG titers in the mouse immune sera, pooled from each group, ranged from 1:8100 to 1:72900 (Figure 1d [↗](#)).

The antigenic profile of N2 neuraminidase

Next, NAI titers were determined in the ferret sera using a panel of H6N2 reassortant viruses in an enzyme-linked lectin assay (ELLA), avoiding interference by anti-H1 HA antibodies in this assay. We generated 27 H6N2 influenza viruses with H6 HA derived from A/mallard/Sweden/2002 and NA derived from a subset of the 44 H3N2 viruses, thereby retaining NA sequence diversity and year of H3N2 virus isolation (Table S2 [↗](#)). The 27 H6N2 viruses could be classified into 4 groups when ordered according to the phylogenetic relatedness of their NA sequence (Figure 2 [↗](#)). The conversion rate of MUNANA into 4-methylumbelliferyl and acetylneuraminic acid by these H6N2 viruses revealed a K_M that ranged from 20.7 to 144.8 μ M as determined by the Michaelis-Menten equation (Figure 2 – figure supplement 1a [↗](#) and table S2 [↗](#)). Interestingly, the K_M differs

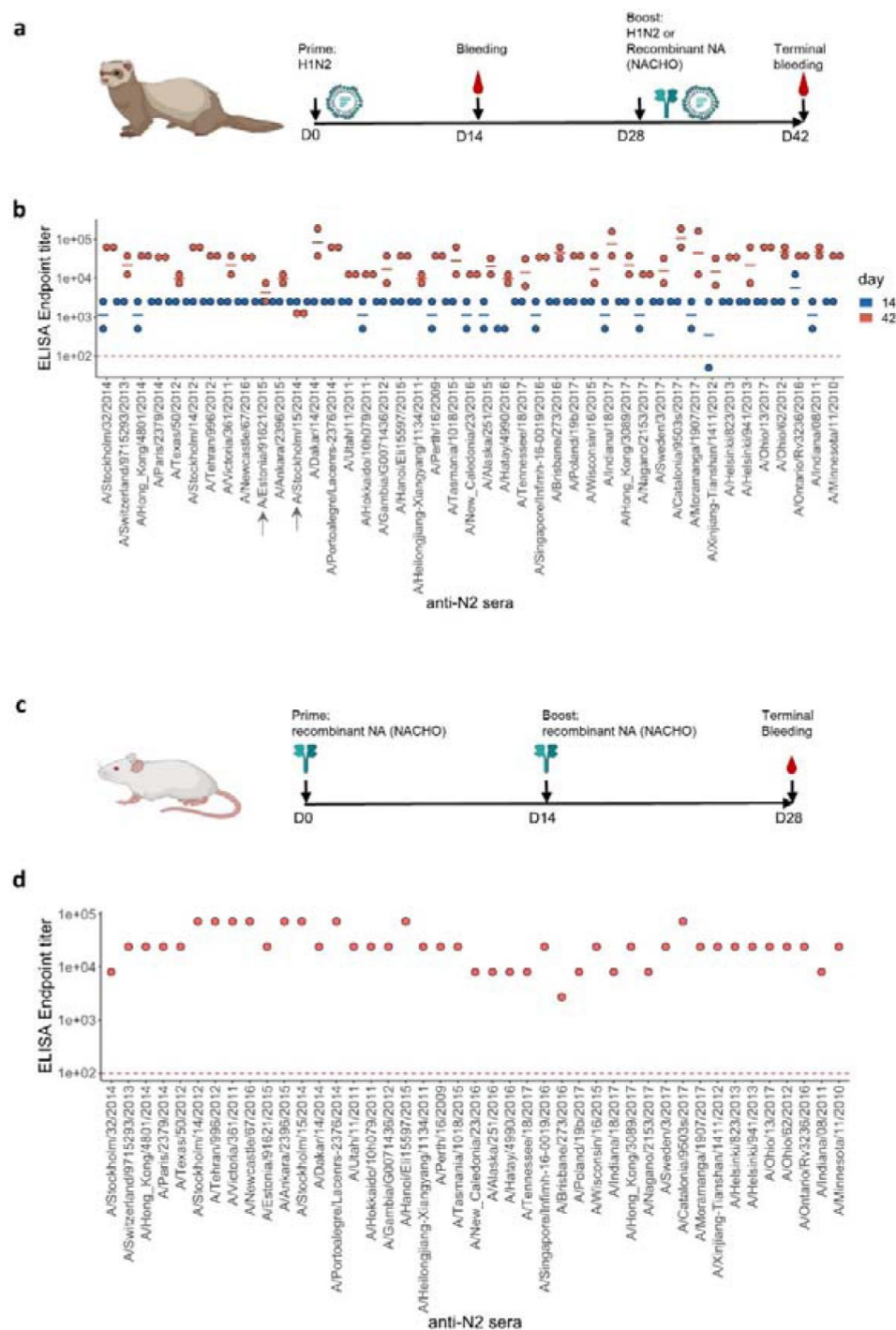


Figure 1.

Generation of anti-N2 sera panel from ferrets and mice.

(a) Ferret immunization and blood sampling scheme. The second immunization on day 28 was performed with homologous NACHO except for the ferrets that had been primed with H1N2_{A/Estonia/91621/2015} or H1N2_{A/Stockholm/15/2014}, which were infected again on day 28 post prime with the same H1N2 viruses. (b) Homologous NACHO ELISA titer of ferret sera sampled on day 14 days after inoculation with H1N2 virus and 2 weeks after the boost with recombinant NA or by re-inoculation with H1N2_{A/Estonia/91621/2015} or H1N2_{A/Stockholm/15/2014} (indicated by arrows on the X-axis). (c) BALB/c immunization and blood sampling scheme. (d) Homologous NACHO serum IgG titers in pooled sera from mice that were immunized twice with recombinant NA. Mouse and ferrets representation were created with *BioRender.com* (<http://biorender.com/>) (2023).

between the distinct phylogenetic groups (**Figure 2 – figure supplement 1a** [↗](#)). There was a negative correlation between the K_M values and the dilution of the H6N2 viruses required to set the 70% maximal NA activity in ELLA (**Figure 2 – figure supplement 1b** [↗](#) and **table S2** [↗](#)).

Using this panel of 27 H6N2s, we determined the NAI titers in the sera obtained from the 44 pairs of ferrets 2 weeks after the boost. ELLAs were performed in duplicate for all serum samples against one H6N2 target at the same time. The reciprocal NAI titers of the ferret sera paired with H6N2 viruses are shown in table S3. In general, the NAI titers of the ferret sera were the highest against homologous and phylogenetically closely related H6N2 viruses.

In **Figure 2** [↗](#), all the tested ferret sera are represented in the rows and the 27 H6N2 viruses against which the sera were evaluated in ELLA are presented in the columns. The serum samples were ordered according to the phylogenetic relatedness of the N2 NA head domain amino acid sequences, showing that these cluster in 4 groups (G1-G4). The ELLA-based IC_{50} NAI titers were plotted relative to the maximum NAI titer obtained against the respective H6N2 target viruses. The heatmap thus represents, for each H6N2 virus, the normalized ELLA IC_{50} titer expressed as $\log_2$ of the fold-difference relative to the serum with the highest IC_{50} value against that H6N2 virus (dark red squares indicate high and blue squares low IC_{50} values) (**Figure 2** [↗](#)).

Next, these normalized NAI values were clustered against each of the H6N2s using the Unweighted Pair Group Method with Arithmetic Mean (UPGMA), which is a distance-based method. This clustering allowed to discern at least 4 major antigenic clusters (AC1 to -4) that are depicted at the top of the heat map (**Figure 2** [↗](#)). It is interesting to note that the H6N2 viruses with NA antigens derived from A/Wisconsin/16/15, A/Alaska/251/15, and A/Kansas/14/2017 do not segregate into AC2, as would be predicted based on the phylogenetic relatedness, but cluster in AC1 based on the UPGMA. However, the overall concordance of the antigenicity with the phylogenetic relatedness, based on the amino acid sequences of the NA head domains, was confirmed by hierarchical clustering with bootstrapping and k-means (**Figure 2 – figure supplements 2a** [↗](#) and **2b**). In line with this, a positive correlation was observed between the pairwise phylogenetic distances and the normalized NAI IC_{50} values for each of the tested H6N2s (**Figure 2 – figure supplement 3** [↗](#)).

We further tested the NAI activity of ferrets' sera obtained 2 weeks after the prime infection against at least one H6N2 virus from each antigenic cluster. These H6N2 viruses were selected based on their antigenic relatedness as deduced from the ferret immune sera analysis (**Figure 2** [↗](#)) with 2 representatives from group 1 (Tex12, and Per09) and 1 each from group 2 (Kan17), group 3 (Hel823), and group 4 (Ind11). The normalized data confirmed the antigenic clustering observed in sera after boost (**Figure 2 – figure supplement 4** [↗](#)).

The NAI titers were also determined in the sera obtained from the 43 groups of NACHO-immunized mice against 7 of the H6N2 viruses. These H6N2 viruses were selected as described above with the addition of HK14 and Swi13 H6N2s, both belonging to antigenic cluster 1. Serum NAI IC_{50} values against a given H6N2 virus were normalized against the IC_{50} of the mouse serum that had the highest NAI titers against the tested H6N2 virus. UPGMA-based clustering showed that the mouse NACHO immune sera displayed a similar recognition pattern as the ferret immune sera that had been generated by H1N2 priming followed by a NACHO boost (**Figure 2** [↗](#)). For example, HK14 H6N2 NA activity is strongly inhibited by sera from group 1, is poorly inhibited by most of the sera from group 2 and is hardly inhibited by group 3 and 4 sera from both ferrets and mice. Ind11 H6N2 NA activity, on the other hand, was inhibited by group 1 and 4, and hardly by group 2 and 3 immune sera. Overall, the normalized NAI IC_{50} values of the ferret and mouse immune sera correlated very well with coefficients values ranging from 0.33 to 0.78 (**Figure 2 – figure supplement 5b** [↗](#)).

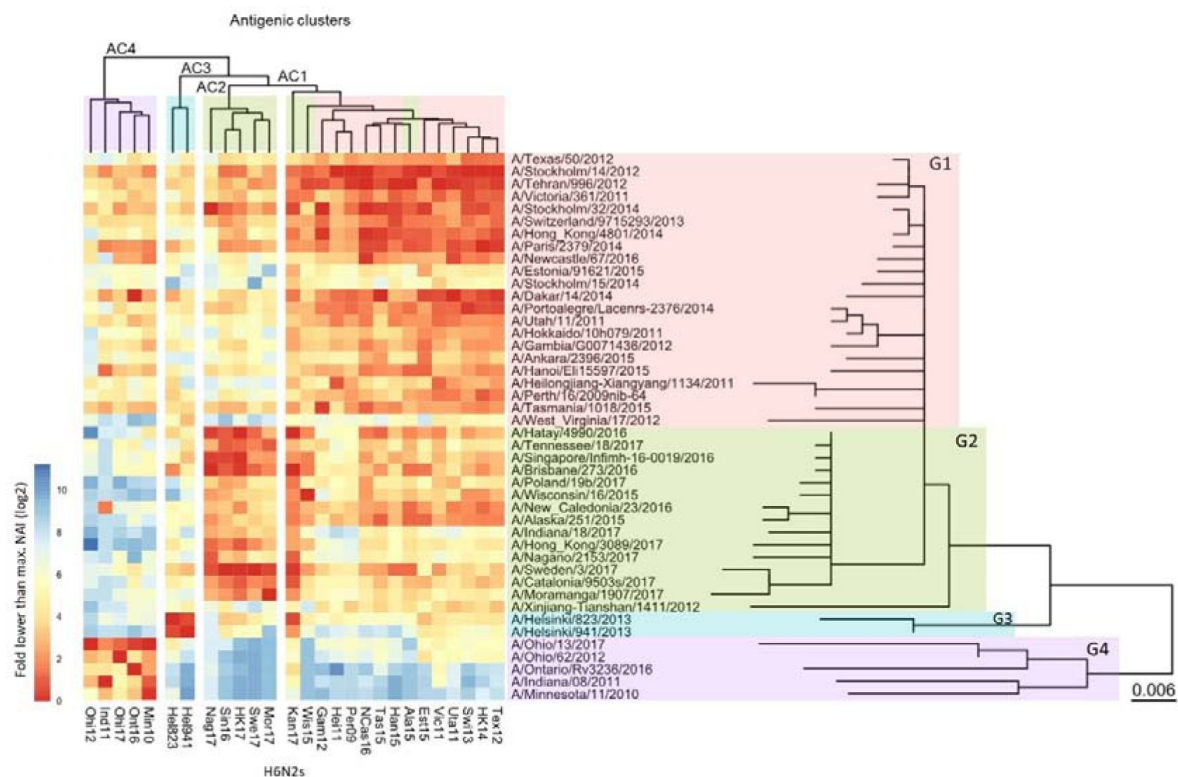


Figure 2.

Breadth of N2 NA inhibition.

Phylogenetic tree based on the N2 NA head domain amino acid sequences and heat-map representing the average of normalized neuraminidase inhibition per H6N2 [$\log_2 (\max \text{NAI}/\text{NAI})$] determined in ferret sera after the boost (listed vertically). The red-to-blue scale indicates high-to-low NAI observed in ELLA against the H6N2 reassortants (listed at the bottom). UPGMA clustering of H6N2s inhibition profiles are shown on top of the heat map and coloured according to the phylogenetic groups.

Association analysis reveals amino acid residue substitutions that strongly impact NAI

NA of human influenza viruses is subject to antigenic drift, predominantly driven by the NAI activity that is detectable in the host serum (Air et al., 1985 [↗](#); Yasuhara et al., 2019 [↗](#)). There are a total of 102 amino acid differences between the head domains of the NAs that were used to generate the 44 ferret immune sera (Figure 3 - Figure supplement 1 [↗](#)). These 102 variable amino acids in the N2 head domain are predominantly exposed on the NA surface (Figure 3 - figure supplement 2 [↗](#)). To try to identify which of these amino acid substitutions have a significant impact on NAI, we performed an association analysis for each of the tested H6N2 viruses. For this analysis, the NAI IC₅₀ value was compared between groups of N2 sera according to the presence or absence of a substitution of any of the 102 amino acid position that are different between the panel of 44 tested N2s against which sera had been raised in ferrets. As a result, a p-value is obtained for each position that reflects its possible impact on NAI (Figure 3a [↗](#)). The significance values higher than 1.3 (p<0.05) of representative strains of the 4 antigenic groups were represented on the surface of N2 structure (Figure 3b [↗](#)). No obvious antigenic patches, however, could be distinguished on the N2 NA surface.

A similar association analysis was performed using the NAI values obtained with the mouse NACHO immune sera. The calculated p-values obtained from the mouse and ferret NAI data were in line with each other (Figure 3 - figure supplement 3 [↗](#)).

Residues close to the catalytic pocket are more likely to impact NAI

We next calculated the distance of every C α atom of each amino acid present in the N2 structure (PDB: 6BR6, from A/Perth/16/2009 NA) to each C α of the variable amino acids. We then determined the correlation of those distances with the p-values that were previously determined in the association analysis. The resulting correlation coefficients (Tau values), indicate a similar pattern among all 27 tested H6N2s (Figure 4a [↗](#)). When visualized on the N2 surface structure, the Tau values revealed a geometrical distribution with the negative correlations being in closer proximity to the catalytic pocket (Figure 4b [↗](#)). Similar data was obtained when mice sera were analyzed this way (Figure 4 - figure supplement 1 [↗](#)).

Prediction of the N2 antigenicity based on a machine learning method

The antigenic drift of human H3 HA has been extensively studied and modelled. (Meyer and Wilke, 2015 [↗](#); Smith et al., 2004 [↗](#); Westgeest et al., 2012 [↗](#); Wu et al., 2020 [↗](#)). To model the NA drift observed in our panel of N2 NAs, we first calculated the antigenic distances between the distinct NA sera, using a similar approach as has been used to visualize the HA antigenic map (Cai et al., 2012 [↗](#)). These antigenic distances can be visualized using classical multidimensional scaling, and the resulting antigenic map (Figure 5 - figure supplement 1a [↗](#)) revealed a distribution that is compatible with the representation of phylogenetic distances (Figure 5 - figure supplement 1b [↗](#)).

Then, using the antigenic distances and a random forest method (RF), a model was built based on amino acid sequence information. A similar model has been used by Yao *et al.* to predict the antigenicity of H3 HA (Yao et al., 2017 [↗](#)). Here, the optimized model was built using a combination of 25 randomly sampled data in 1500 decision trees. The resulting model generates predictive antigenic distances that strongly correlate with the observed values (Figure 5a [↗](#)). The RF also enables scoring of amino acid residues according to their importance in the decision trees. One measure of importance can be defined as how early a change in a position is placed in the generated decision trees (mean minimal depth). Another measure of importance is how much the

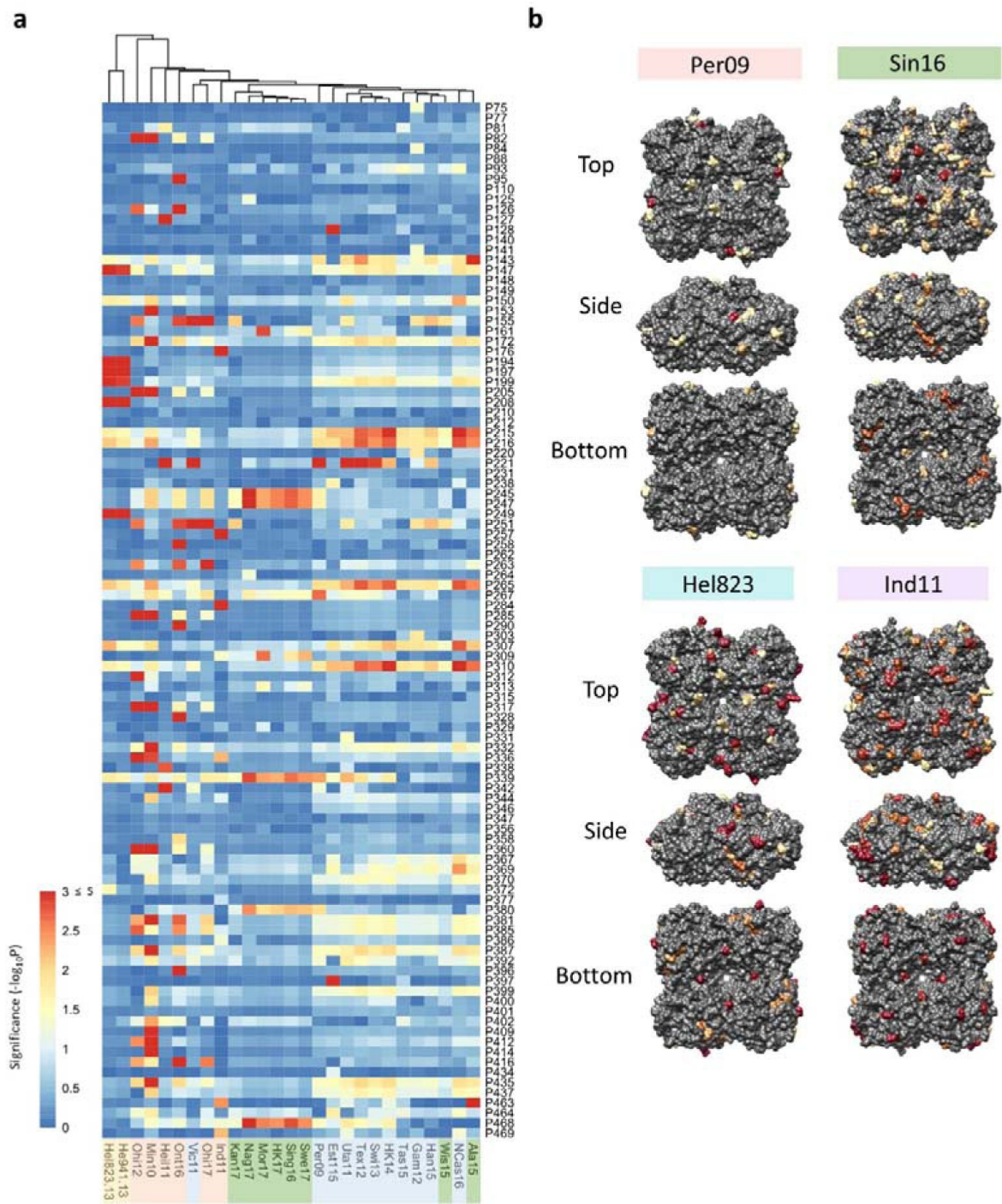


Figure 3.

Association analysis reveals impact of amino acid mutations on NAI.

(a) Association study performed for each H6N2 virus (listed at the bottom) to determine the probability of each variable amino acid (listed on the right) present in the panel of 44 NAs to impact the NAI. The red-to-blue scale represents significance values ($S = -\log_{10}P$) associated with each amino acid substitution. The dark red in the color scale represent all values that are equal or higher than 3 (corresponding to p -values ≤ 0.001). The colour shades on the bottom side indicate the distinct phylogenetic groups. (b) The significant values ($p < 0.05$) were represented in the surface of N2 tetramer (PDB accession number 4H53) for one representative strain of each phylogenetic group with a yellow to red scale of significance ($-\log_{10}P$).

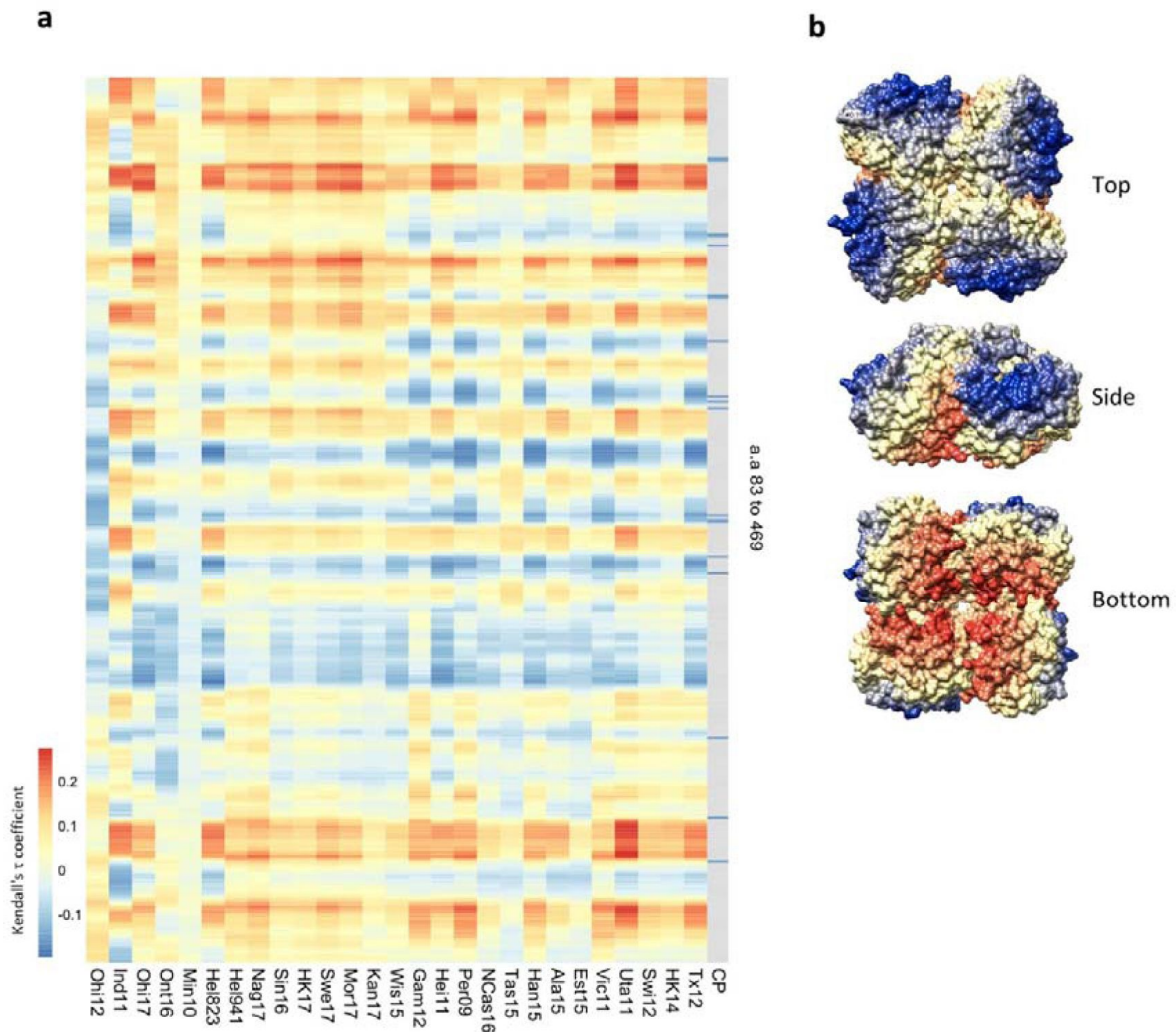


Figure 4.

Residues close to the catalytic pocket are more likely to impact NAI.

(a) Significance values (102 variable amino acids) obtained from the association study were correlated to the distance between every amino acid and each of the variable amino acids in the NA structure. The red-to-blue scale represents the Tau values obtained in the Kendall correlation. Amino acids 83 to 469 are shown from top to bottom in the heatmap. The H6N2s used in the NAI panel are listed at the bottom. The grey column (catalytic pocket (CP)) indicates the residues that are part of the catalytic pocket that directly interact with substrate (R118, D151, R152, R224, E276, R292, R371, and Y406) and the residues that play a key role in stabilizing the catalytic site (E119, R156, W178, S179, D198, I222, E227, H274, E277, N294, and E425). (b) Tau values obtained from Tex12 H6N2 correlation are represented on the surface of the N2 structure of A/R/5+/1957 (PBD 4H53).

standard error mean increases when a variable is permuted. Finally, the prevalence of a variable amino acid in nodes of the decision tree can also indicate its importance. Those importance scores were represented in a multi-way plot (**Figure 5b** [↗](#)). The importance scores highlighted similar positions as previously determined in the association study. Finally, we used NA sequences of recent seasonal influenza vaccine recommended H3N2 strains to predict their antigenic distance relative to our panel of N2s, including 3 recent N2 NA components (from A/HongKong/45/2019, A/Tasmania/503/2020, and A/Darwin/6/2021) that were not present in our data set. The predicted antigenic distances were plotted in the heatmap in a red to blue scale, indicating high to low antigenic distance, respectively (**Figure 5c** [↗](#)). The RF clearly reproduced the drift observed in AC2 versus AC1. The modelling also indicates that the antigenic distance of the 3 recent N2s to their corresponding group 2 NAs is increasing (represented by the dark blue turning light blue, from the A/Singapore/infimh-16-0019/2016 towards the A/Darwin/6/2021 in G2) whereas their distance to group 4 NAs is decreasing (dark red becoming yellow from the A/Singapore/infimh-16-0019/2016 towards the A/Darwin/6/2021 in G4).

Discussion

The sequence diversity of N2 protein derived from the 44 selected human H3N2 viruses isolated between 2009 and 2017 allows to classify them in 4 major phylogenetic groups. Group 1 contains 22 N2 NAs, 4 of which were derived from vaccine recommended H3N2 strains: A/Perth/16/2009, A/Texas/50/2012, A/Switzerland/9715293/2013, and A/Hong Kong/4801/2014. Phylogenetic group 2 comprises 15 N2 NAs, one of which is derived from a vaccine H3N2 recommended strain: A/Singapore/INFIMH-16-0019/2016. The third phylogenetic group contains two NAs that are derived from H3N2 strains that had been isolated in Helsinki in 2013. Remarkably, the NA sequence of these two strains is closely related to that of H3N2s that circulated more than 15 years earlier (e.g. 99% identity with NA of A/Sidney/5/1997). The fourth phylogenetic group is composed of NAs derived from zoonotic cases with swine H3N2 strains isolated between 2012 and 2017. Swine-to-human transmission of H3N2 viruses have occurred multiple times in the USA in 2012 and continue to be detected at low frequency until to date ([CDC, 2023](#) [↗](#); [Elbe and Buckland-Merrett, 2017](#) [↗](#)).

We assessed the antigenic diversity of the NA panel by generating 27 recombinant H6N2 viruses and 44 ferret NA immune sera produced by H1N2 priming followed by boosting with homologous recombinant NA (NACHO; n = 42) or by reinfection with H1N2 (A/Estonia/91621/2015 and A/Stockholm/15/2014). Most of the ferrets had detectable serum anti-N2 antibodies by day 14 after infection with H1N2. Anti-NA antibody titers increased after a boost with non-adjuvanted homologous recombinant CHO-produced soluble tetrameric NA suggesting that the use of recombinant NA may boost naturally acquired anti-NA antibody responses in the human population.

The NAI activity of the ferret sera against 27 tested H6N2s reassortants revealed a pattern of inhibition that largely coincides with the phylogenetic distances based on the NA head domain sequences. In the tested panel of H6N2s, at least 4 major antigenic groups were identified. We noted that H6N2 with Kan17 NA had a broader profile of inhibition than other H6N2 viruses from group 2, being also inhibited by serum raised against NA from phylogenetic groups 3 and, to a lesser extent, group 4. Additionally, the H6N2s Wis15 and Ala15 have an antigenic behaviour that is more closely related to antigenic cluster 1, despite belonging to group 2 according to their phylogenetic similarities. Based on the NAI panel, an association analysis was performed to try to identify amino acid changes that affect NAI. It is interesting to note that the amino acid substitutions S245N and S247T, combined with substitution P468H, were identified in this association analysis. Those substitutions were described as major contributors of the NA antigenic drift that occurred in H3N2 in 2014 ([Wan et al., 2019](#) [↗](#)). NAs from A/Wisconsin/16/2015 and A/Alaska/251/2015 also have the S245N and S247T substitutions and phylogenetically belong to

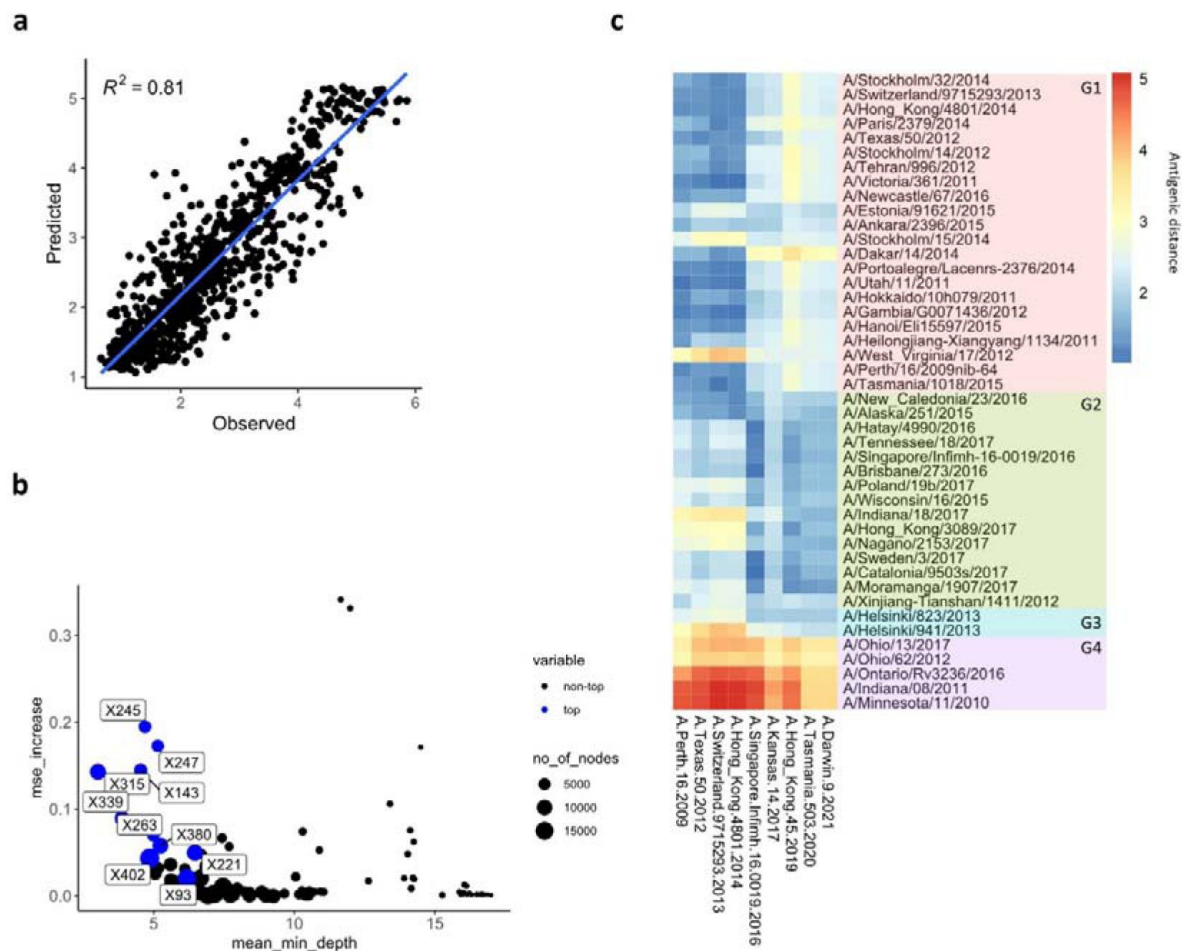


Figure 5.

Prediction of N2 antigenicity based on machine learning.

(a) Predicted vs observed normalized NAIs are shown. (b) Multi-way importance plot indicating the increase in mean standard error, the mean depth, and number of nodes. The top 10 most important amino acid changes are highlighted in blue and annotated with their position in the NA sequence. (c) The heatmap indicates the predicted antigenic distances of N2s from recent H3N2 isolates (at the bottom of the heatmap) relative to the N2s that were used in the NAI breadth panel (listed vertically). The antigenic distances were calculated based on NA protein sequence.

group 2 NAs. However, those 2 NAs lack the P468H mutation and are antigenically closer to group 1 NAs, confirming the requirement of P468H combined with the S245N and S247T substitutions for this NA antigenic drift (Wan et al., 2019 [DOI](#)). This observation suggests that NA antigenic drift may sometimes require a combination of amino acid changes rather than single mutations. Wang and colleagues have very elegantly demonstrated that N2 antigenic evolution is constrained by charge balances in an antigenic region, which can partially explain the observation that NA amino acids can undergo co-evolution to result in antigenic drift (Wang et al., 2021 [DOI](#)). Interestingly, NAs originated from H3N2 variants classified in phylogenetic group 4 were antigenically closer to N2s belonging to phylogenetic group 1. This suggests that the antigenic drift of recent NAs from group 2 increases the antigenic distance from the N2s of swine origin.

Noteworthy, the analysis of the sera obtained from ferrets after infection shows that, despite the lower homologous anti-NA titers, the inhibition pattern evaluated against 5 distinct H6N2s (composed of at least one representative of each antigenic group) is in accordance with the response obtained after boosting with recombinant protein. This suggests that a boost with homologous recombinant soluble tetrameric NA does not alter the breadth of the antibody response induced by H1N2 infection only.

NAI was also determined against 7 distinct H6N2 reassortant viruses using a panel of 43 anti-N2 NA sera raised in mice by immunization with adjuvanted NACHOs. The observed NAI responses again allowed clustering of the N2 NAs into 4 main antigenic groups, which correlates with data obtained in ferrets. Studies in mice can thus offer a convenient alternative to the use of ferrets in future NA antigenicity studies. This data also supports the hypothesis that there may be universal rules of immunogen antigenicity and immunogenicity (Altman et al., 2015 [DOI](#)).

The determined K_M values of the H6N2 viruses used here, indicates that substrate affinity tends to vary according to the phylogenetic relatedness of N2s. The N-glycosylation at position 245 was previously described to be responsible for reducing N2 affinity for its substrate (Powell and Pekosz, 2020 [DOI](#)). Unlike group 2, which is primarily characterized by the presence of a N-glycan at position 245, NAs from phylogenetic group 4 have a low K_M , indicating an increased affinity for substrate. This result suggests that phylogenetic drift can be accompanied by changes in NA enzymatic properties. Curiously, the Ala15, Wis15, and Kan17, despite having antigenic properties closer to group 1, still have K_M values that are comparable to those of group 2 N2 NAs.

After the determination of NA antigenic distances, we used random forest modelling to estimate the impact of amino acid changes on antigenic distances. We obtained a strong correlation of predicted versus observed antigenic distances. However, it is important to note that the training dataset was based on the 102 mutations present in our panel, limiting the predictions for other positions. Furthermore, random forest enabled the classification of the variable amino acids according to importance scores. It is important to highlight that co-occurring mutations in our panel (the ones present in main branches of the phylogenetic tree) cannot be individually assessed by association analysis or the random forest model. The individual weight of those mutations on NA drift remains to be experimentally demonstrated. Interestingly the modelling could put in evidence the gradual drift of recent N2s which were not present in the dataset (A/Hong Kong/45/2019, A/Tasmania/503/2020, and A/Darwin/6/2021).

The protective potential of NA antigen against influenza has been underscored by multiple studies. It is likely that NA will be included as a standard component of next generation influenza vaccines. The difficulties of producing stable tetrameric NA may be overcome in the future thanks to protein design or with the advent of mRNA-based vaccines (Catani et al., 2022 [DOI](#); Chivukula et al., 2021 [DOI](#); Ellis et al., 2022 [DOI](#)). The antigenic modelling associated with sequence surveillance may increase our predictive ability, highlighting major potential drifts and support NA antigen designs that may elicit broader NAI responses compared to natural NA.

In summary, our study shows that immunization with recombinant N2 NA immunogen can boost anti-NA serum antibody responses in ferrets that were primed by infection with an influenza virus that carries the homologous NA. Prime by infection allowed to use non-adjuvanted NACHO in the subsequent boost. At least 4 major antigenic groups could be distinguished in N2 NA derived from human H3N2 viruses that were isolated between 2009 to 2017, and those antigenic groups predominantly correspond with the phylogenetic relatedness of the N2 head domains. In addition, the amino acid substitutions in close proximity of the NA catalytic pocket are more likely to affect NAI patterns.

Material and Methods

Rescue of H1N2 and H6N2 reassortants

All reassortant H1N2 and H6N2 viruses included in the study were generated by reverse genetics, with each reassortant expressing the targeted NA antigen, internal genes from A/Puerto Rico/8/1934 H1N1 (PR8), and the HA from PR8 H1N1 or A/mallard/Sweden/81/2002 H6N1, respectively. Briefly, HA and NA segments including non-coding regions were generated by custom gene synthesis (Geneart AG), while PR8 internal segments were derived from a viral isolate as previously published (Marsh and Tannock, 2005 [↗](#)). All segments were cloned into a bi-directional transcription plasmid derived from pUC57 (Genscript) including polymerase (Pol) I and Pol II promoters, as previously described (Hoffmann et al., 2000 [↗](#)), and the entire set of 8 plasmids was used to transfect 293FT cells (Thermo Fisher Scientific) using Lipofectamine 2000 CD (Thermo Fisher Scientific). Twenty-four hours after transfection, Madin-Darby Canine Kidney Cells (MDCK-) ATL cells (ATCC) were added to the transfected cells in the presence of TPCK-treated trypsin (Sigma) to allow influenza virus propagation. Cell culture supernatants containing influenza virus were harvested 7 days post-MDCK addition and blindly passaged in 8–10-day-old embryonated chicken eggs (Charles River Laboratories, Inc.). Inoculated eggs were incubated at 37°C for 48 h, then cooled to 4°C for 12 h, prior to allantoic fluid harvest and clarification by low-speed centrifugation (3,000 rpm, 20 min). For H6N2 viruses only, high-yield stocks were generated by an additional passage in eggs as described above. Virus titers were determined by plaque assay on MDCK cells.

Design and production of recombinant proteins

The coding information of the NACHO proteins was cloned under the transcriptional control of the CMV promoter in the pCDNA3.4 plasmid with a CD5 secretion signal, an amino-terminal His-tag and the thrombin cleavage signal followed by tetrabrachion-NA. NACHOs were expressed in a mammalian cell culture system as previously described with modifications (Catani et al., 2022 [↗](#)).

Animal experiments

The ferret immunization experiment was conducted at Bioqual, Inc., MD (USA). The experiments were approved by the Institutional Animal Care and Use Committee of the Bioqual (reference number 17-021.6). Outbred 18–22 weeks old naïve male and female Fitch ferrets were obtained from Triple F, PA (USA) and housed under specified pathogen-free conditions with food and water *ad libitum*. Ferrets were primed on day 0 by intranasal challenge with influenza H1N2 reassortant virus (1,000 µl/dose, split evenly between nostrils). Four weeks after initial infection (day 28), ferrets were immunized with 45 µg of NACHO (500 µl/dose, intramuscularly, split evenly between hind thighs) corresponding to homologous NA used in day 0 prime. Alternatively, and due to low yield of selected recombinant NACHOs, ferrets were boosted on day 28 with 15 µg of homologous NACHO (A/Perth/16/2009nib-64, A/Tasmania/1018/2015, A/Utah/11/2011, and A/Virginia/17/2012) or were inoculated with the same H1N2 reassortant virus used in the prime

(A/Estonia/91621/2015 and A/Stockholm/15/2014). Sera from bleeds two weeks after the prime (day 14) and booster vaccination (day 42) were tested for immunogenicity and seroconversion in serological assays.

The mouse immunization experiments were conducted according to the Belgian legislation (Belgian Law 14/08/1986 and Belgium Royal Decree 06/04/2010) and European legislation on protection of animals used for scientific purposes (EU directives 2010/63/EU and 86/609/EEC). Experimental protocols were all approved by the Ethics Committee of the Vlaams Instituut voor Biotechnologie (VIB), Ghent University, Faculty of Science (EC2019-035). Female BALB/c mice, aged 6–7 weeks, were purchased from Charles River. The mice were housed in a specified pathogen-free animal house with food and water *ad libitum*. Animals were immunized intramuscularly in the right quadriceps. Immunization was performed with 50 µl containing 1 µg of recombinant protein (tetra-brachion stabilized NAs, referred to as NACHOs). All, except A/West Virginia/17/2012 NACHOs were used as immunogens in mice. All immunizations were adjuvanted with a 1:1 volume of AF03 (25 µl of antigen in PBS + 25 µl of AF03 per dose) (Caillet et al., 2010 [\[10\]](#)).

Two or three weeks after the boost immunization, mice were humanely sacrificed with an anaesthetic overdose of Nembutal (6 mg per mouse) and bled retro-orbitally. The obtained blood samples were incubated for 30 min at 37°C to allow clotting, which was followed by centrifugation at 10,500 g for 5 minutes. The supernatant (the serum) was recovered and submitted to a second centrifugation at 10,500 g for 5 minutes. Cleared sera were stored at -20°C before use in serological assays.

Elisa

Anti-tetNA IgG levels in mouse and ferret sera were determined by capture ELISA using tetNAs in Pierce™ nickel-coated plates (cat. # 15442). Recombinant proteins were diluted in DPBS (Life technologies cat. # 14040-182) (tetNA was diluted to 0.5 µg/ml). Then, 50 µl of the coating antigen solutions was added to each well and the plate was incubated at room temperature for 1 h on a shaking platform. The wells of the plates were then washed 3 times with PBS-T (Sigma cat. # P3563-10PAK) and blocked for 1h with 1% BSA (for the mouse sera) or 10% milk (for the ferret sera) in DPBS. After blocking, the wells of the plates were washed once with PBS-T and incubated with a 3-fold serial dilution, starting at a 1/100 dilution, of serum in DPBS, 0.5% BSA, 0.05% tween-20 for 2h at room temperature on a shaking platform. The wells of the plates were then washed 5 times with PBS-T and incubated with a 1:5000 dilution of anti-mouse IgG-HRP (GE healthcare cat. # NA931-1ml), or with a 1:100000 dilution of anti-ferret IgG-HRP (Abcam cat. # 112770) in DPBS, 0.5% BSA, 0.05% Tween-20. The 3,3',5,5'-tetramethylbenzidine (TMB) substrate (BD cat. # 555214) was added after 3 washes with PBS-T and the reaction was stopped after 5 min by addition of 50 µl of 1 M H₂SO₄. The optical density (O.D.) in each well was determined at 450 nm and, as a reference, 655 nm using an iMark™ Microplate Absorbance Reader (Bio-rad). The end point titer was determined for each serum sample by scoring the dilution that resulted in an O.D. that was equal to or 2 times higher than the background O.D. obtained from pre-immune control sera dilution series.

Enzyme-linked lectin assay to determine neuraminidase inhibition titers

Fetuin (Sigma cat. # F3385) was diluted into coating buffer (KPL cat. # 50-84-01) to a concentration of 25 µg/ml and 50 µl was added to the wells of Nunc MaxiSorp™ plates (Thermofisher cat. # 44-2404-21), which were incubated overnight at 4°C. The coated plates were then washed 3 times with PBS-T (Sigma cat. # P3563-10PAK) and incubated overnight with 60 µl of a dilution of H6N2 and 60 µl of 2-fold serial dilution of serum, starting at a 1/20 dilution, in sample buffer (1X MES VWR cat. # AAJ61979-AP: 20mM CaCl₂, 1% BSA, 0.5% Tween-20). These dilutions of the H6Nx viruses correspond to the 70% maximum activity of NA from the respective viruses as determined in the ELLA assay. Fetuin-coated plates were then washed 3 times with PBS-T and incubated for 1 h with

a solution of PNA-HRP (cat. # L6135-1MG, Sigma) at 5 µg/ml in conjugate diluent (MES pH 6.5, 20 mM CaCl₂, 1% BSA). The plates were washed 3 times with PBS-T, TMB substrate was added and then the plates were incubated for 5 minutes before the reaction was stopped by the addition of 50 µl of 1 M H₂SO₄. The optical density was measured at 450 nm and, as a reference, 655 nm in an iMark™ Microplate Absorbance Reader (Bio-rad). Half maximum inhibitory concentrations (IC₅₀) values were determined by non-linear regression analysis (GraphPad Prism software).

NA Enzyme Kinetics Assay

The enzymatic activity of NACHOs and H6N2s was determined with the fluorogenic small substrate 4-methylumbelliferyl-α-d-N-acetylneuraminic acid (MUNANA) as previously described (Marathe et al., 2013). First H6N2 dilutions were tested to ensure that MUNANA conversion into 4-methylumbelliferone (4-MU) generated fluorescence over time at a linear ratio. Ten microliters of 1 mM of MUNANA (Sigma cat # M8639) in 200 mM sodium acetate buffer (pH 6.5) containing 2 mM CaCl₂ and 1% butanol was incubated with 40 µl of PBS solution containing viral dilution. Conversion to 4-MU was monitored every 2 min for 1 h using a BMG Fluostar OPTIMA reader (excitation at 365 nm and emission determined at 450 nm). Second, using the selected dilution of virus, the assay was repeated with dilutions of MUNANA ranging from 2.5 to 100 µM. A standard curve with 800 to 25 pmoles/well of 4-MU (Sigma cat # M1508) was used to extrapolate the molar conversion of MUNANA. K_M was calculated using GraphPad Prism version 8.4.3.

Association study

For the association analysis, under the assumption of homogeneous genetic relatedness among the selected H3N2 virus strains, a simple linear regression model for mutation-trait association analysis was fitted to the data for each H6N2 separately. The significance values were calculated by comparing IC₅₀ values of sera that were classified into 2 groups according to the presence or absence of variation in the amino acid of reference Tex12 sequence.

Correlations with distances from the NA catalytic pocket

Tau values were determined by correlating distances with significance values obtained from the association analysis or the importance scores obtained from machine learning. A similar method was described by Meyer et al. to define geometrical constraints of HA evolution (Meyer and Wilke, 2015). Briefly, the relative distances of amino acids were calculated based on the structure of A/Perth/2009 (H3N2) NA (PDB: 6BR6) (Hadházi et al., 2018). The distances were calculated from each Ca to every other Ca. The distances from every residue to the 98 variable residues that are present in the panel of N2s and 6BR6 structure were then correlated with the significance values that were obtained in the association analysis, or importance scores obtained in the machine learning. Correlation coefficients were visualized on the structure of N2 NA derived from pandemic A/RI/5+/1957 H2N2 virus (PDB: 4H53) (Vavricka et al., 2013) using Chimera 1.14.

Random Forest algorithm

The Random Forest algorithm was adapted from (Yao et al., 2017). Briefly, the non-conserved positions in the multiple sequence alignment of N2 head domain amino acid sequences of all tested N2s were selected. The feature matrix (X) was created by assessing matches/mismatches on each non-conserved position for each pair of N2 head domain amino acid sequences as:

$$X = \begin{matrix} & & X_1 & X_2 & \dots & X_m \\ \begin{matrix} V_1 V_2 \\ \vdots \\ V_1 V_n \\ \vdots \\ V_{n-1} V_n \end{matrix} & \begin{matrix} 0 & 1 & \dots & 1 \\ \vdots & \vdots & \dots & \vdots \\ 1 & 0 & \dots & 1 \\ \vdots & \vdots & \dots & \vdots \\ 1 & 0 & \dots & 1 \end{matrix} \end{matrix}$$

where X_i represents the non-conserved positions in the multiple sequence alignment m is the number of non-conserved positions (102 in total)

V_i represents the N2 head domain sequence for each tested NA

n is the number of tested sera (44 in total)

0 represents a match, 1 represents a mismatch in the alignment

The response vector (Y) is composed of the antigenic distances for each pair of tested sera. These antigenic distances were determined based on NAI, similarly to the method used to determine HI based immunogenic distances that was described by Cai *et al.* In summary, the distances between sera was calculated as $\frac{1}{n} \sum_{t=1}^n |d[t, i] - d[i, t]|$, which represent average distances between 2 serum samples among all tested H6N2s (Cai *et al.*, 2012 [DOI](#)). We applied the random forest function from the randomForest package in R to construct the model. We set the bootstrapping (tree) number to 1500 and the number of features to 25 for each tree. Accuracy was determined as the root-mean-square error (RMSE), calculated using the RMSE function from the yardstick package in R.

Additionally, predicted antigenic distances of the out-of-bag samples were compared to the observed distances by fitting a linear model and calculating the R^2 -value. To calculate the importance of each feature (X_i), the following metrics were calculated using the measure importance function of the randomForestExplainer package in R:

- - mean minimal depth: mean of the depths of the node in each tree that splits on this feature and is the closest to the root of the tree (minimal depth). The lower, the more important the feature since a low depth means that many observations are divided into groups based on this feature.
- - mse decrease: mean decrease of accuracy in predictions on out of bag samples when the feature is excluded from the model. The higher, the more important the feature.
- - number of nodes: total number of nodes in the forest that use the feature for splitting. The higher, the more important the feature.

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Competing interests

XS reports grants from Sanofi Vaccines. GCG, TS, RG, JH, JC, IVU, TF, SS, SR, LR, and TUV are or were Sanofi employees and may hold shares and/or stock options in the company.

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Author Contributions

JPPC, TY, AS, IVU, TF, RG, JH, SS, SR, LL and JC performed experimental work. JPPC, TY, AS, RG, GCG, TUV, and XS designed the experiments, analyzed and interpreted the data. JM and MV analyzed the data. JPPC and XS wrote the manuscript and are accountable for accuracy and integrity of the work. All authors proofread and critically reviewed the manuscript.

H3N2 strain NA source	Specific Activity nmole/min.µg
A/Stockholm/32/2014	9,62
A/Switzerland/9715293/2013	6,52
A/Hong Kong/4801/2014	4,13
A/Paris/2379/2014	2,42
A/Texas/50/2012	5,21
A/Stockholm/14/2012	2,42
A/Tehran/996/2012	8,46
A/Victoria/361/2011	2,71
A/New_Castle/67/2016	3,22
A/Estonia/91621/2015	5,3
A/Ankara/2396/2015	2,62
A/Stockholm/15/2014	1,14
A/Dakar/14/2014	2,37
A/PortoAlegre/Lacnrs-2376/2014	2,55
A/Utah/11/2011	1,40
A/Hokkaido/IOH079/2011	0,84
A/Gambia/2012	1,60
A/Hanoi/Eli15597/2015	3,29
A/XinJiang-Tianshan/1411/2012	0,27
A/Perth/16/2009nib-64	5,3
A/Tasmania/1018/2015	27,48
A/NewCaledonia/23/2016	8,55
A/Alaska/251/2015	4,29
A/Hatay/4990/2016	5,83
A/Tennessee/18/2017	5,99
A/Singapore/Infimh160019/2016	8,01
A/Brisbane/273/2016	6,94
A/Poland/19b/2017	5,81
A/Wisconsin/16/2015	1,40
A/Indiana/08/2011	4,13
A/Hong-Kong/3089/2017	6,17
A/Nagano/2153/2017	2,29
A/Sweden/3/2017	2,76
A/Catalonia/9503s/2017	1,31
A/Moramanga/1907/2017	2,09
A/Heilongjiang_Xiangyan/1134/2011	6,61
A/Helsinki/823/2013	6,53
A/Helsinki/941/2013	7,02
A/Ohio/13/2017	2,59

Table S1.

Specific activity of the recombinant NACHOs.

A/Ohio/62/2012	4,28
A/Ontario/RV3236/2016	3,89
A/Indiana/18/2017	4,27
A/Minnesota/11/2010	4,25

Table S1. (continued)

H6Nx reassortant(*)	Titer (PFU/ml)	dilution to 70% max activity (ELLA)	K _M (MUNANA)
A/Ohio/62/2012 (Ohi12)	2.00E+08	2269.7	20.7
A/Indiana/08/2011 (Ind11)	2.50E+08	750.9	40.1
A/Ohio/13/2017 (Ohi17)	3.00E+08	521.5	33.5
A/Ontario/Rv3236/2016 (Ont16)	1.75E+08	504.1	31.5
A/Minnesota/11/2010 (Min10)	2.00E+08	1031.4	41.3
A/Helsinki/823/2013 (Hel823)	1.50E+08	155.2	45.9
A/Helsinki/941/2013 (Hel941)	3.50E+08	341.6	45.3
A/Nagano/2153/2017 (Nag17)	2.50E+08	213.6	97.8
A/Singapore/Infimh-16-0019/2016 (Sin16)	6.50E+07	257.1	93.3
A/Hong_Kong/3089/2017 (HK17)	2.00E+08	301.6	85.2
A/Sweden/3/2017 (Swe17)	6.00E+08	97	71.5
A/Moramanga/1907/2017 (Mor17)	1.40E+08	151.4	56.8
A/Kansas/14/2017 (Kan17)	2.50E+08	249	108.7
A/Wisconsin/16/2015 (Wis15)	2.00E+08	63.3	79.6
A/Gambia/G0071436/2012 (Gam12)	2.50E+08	331.5	63.6
A/Heilongjiang-Xiangyang/1134/2011 (Hei11)	2.10E+08	195.6	31.1
A/Perth/16/2009 (Per09)	3.20E+08	630.8	56.4
A/Newcastle/67/2016 (NCas16)	1.50E+08	150.4	48.5
A/Tasmania/1018/2015 (Tas15)	1.70E+08	139.5	54.8
A/Hanoi/Eli15597/2015 (Han15)	1.40E+08	94.6	82.2
A/Alaska/251/2015 (Ala15)	4.50E+08	93.7	144.8
A/Estonia/91621/2015 (Est15)	2.00E+08	281.9	62.1
A/Victoria/361/2011 (Vic11)	3.00E+08	570.3	51.1
A/Utah/11/2011 (Uta11)	2.50E+08	356.3	51.9
A/Switzerland/9715293/2013 (Swi13)	2.70E+08	160.8	64
A/Hong Kong/4801/2014 (HK14)	4.00E+07	106.3	54.7
A/Texas/50/2012 (Tex12)	2.00E+08	276.6	45.1

*Name of the virus from which the N2 NA in H6N2 reassortant virus was derived. The short name as used in the manuscript text is mentioned between brackets.

Table S2.

H6N2 reassortant viruses used in the ELLA assay.

Sera	Swi13	HK14	Tex12	Vic11	Ncas16	Est15	Uta11	Gam12	Han15	Hei11	Per09	Tas15	Ala15
A/Switzerland/9715293/2013	9333.0	6833.5	10301.6	2442.3	1489.8	5530.4	1460.6	3489.9	5253.8	1510.9	1899.6	1649.9	1812.3
A/Hong_Kong/4801/2014	7527.5	10566.4	14983.0	2559.1	4579.4	7944.5	3590.3	7065.0	4798.3	1971.2	2522.9	3842.2	4980.9
A/Texas/50/2012	10751.3	10397.6	8843.5	2089.6	740.9	3821.0	902.1	1519.1	1453.1	1112.9	2506.5	835.8	1490.8
A/Victoria/361/2011(Iv-165)	16804.7	11568.3	11892.0	3863.7	1868.9	3480.4	2250.0	670.9	3963.3	1379.1	2888.7	883.1	1531.9
A/Newcastle/67/2016	7647.2	9702.6	5244.0	1056.9	4218.8	2295.9	5477.0	821.2	1515.9	1254.3	1304.0	1253.1	1021.6
A/Estonia/91621/2015	1277.1	1379.5	911.1	387.1	290.5	2832.9	277.9	335.3	225.6	272.5	193.3	217.2	70.9
A/Utah/11/2011	6121.8	6591.4	6253.2	3392.5	1570.0	3727.5	2893.7	1294.7	2496.3	4050.7	2196.7	376.0	869.2
A/Gambia/G0071436/2012	5513.7	6192.0	5040.0	1226.5	1492.3	6342.9	772.9	730.2	863.5	1810.2	1399.4	836.9	888.0
A/Hanoi/Eli15597/2015	4635.2	8150.5	6232.1	872.7	1617.0	7192.3	1023.4	1343.3	1822.0	1969.9	1168.3	1307.0	2383.4
A/Heilongjiang-Xiangyang/1134/2011	3506.3	3480.2	5123.0	3514.3	356.0	976.7	1657.1	245.9	557.9	7094.8	2056.7	168.3	658.3
A/Perth/16/2009nib-64	4961.8	13872.0	8102.7	2771.2	630.3	3502.5	2644.5	1079.5	441.6	1663.8	2458.4	418.4	748.3
A/Tasmania/1018/2015	6503.6	7894.2	5204.6	1298.1	2335.3	2927.7	1349.1	6451.6	2374.2	709.1	2659.2	1150.5	977.3
A/Alaska/251/2015	7291.7	8216.5	7274.8	1247.6	1050.4	7445.9	1975.1	348.6	1070.2	1828.5	1142.7	2300.4	3418.1
A/Singapore/Infimh-16-0019/2016	2564.5	3869.9	3206.3	1119.2	1038.5	732.7	545.2	252.3	1308.7	196.5	328.1	399.6	778.4
A/Wisconsin/16/2015	6157.8	6249.7	3088.6	693.4	998.2	1259.1	383.6	73.1	595.7	362.7	415.4	442.8	723.5
A/Hong_Kong/3089/2017	713.6	1372.5	1290.5	391.8	312.4	670.9	231.0	144.6	161.5	143.3	88.2	317.5	702.9
A/Nagano/2153/2017	784.6	798.8	567.6	102.6	215.5	530.1	158.5	97.5	182.8	145.2	108.7	172.4	332.8
A/Sweden/3/2017	2250.6	1249.6	2688.6	1818.4	785.2	443.3	632.2	726.1	3344.3	553.2	512.0	910.7	2153.6
A/Moramanga/1907/2017	1126.6	1371.0	720.8	205.3	185.9	1073.8	339.2	212.5	335.6	228.0	201.0	396.1	603.7
A/Helsinki/823/2013	1130.4	1408.4	1382.9	491.8	77.1	1930.9	324.0	141.1	133.6	431.4	289.4	90.0	106.9
A/Helsinki/941/2013	391.1	379.7	662.2	345.4	19.2	837.0	227.2	32.7	25.7	72.6	111.1	23.7	58.9
A/Ohio/13/2017	396.6	554.6	500.2	247.4	56.6	377.9	185.1	112.0	75.2	168.4	219.8	80.1	10.0
A/Ohio/62/2012	445.7	644.5	694.0	157.0	33.1	1024.6	207.9	37.9	77.9	98.5	152.6	86.9	22.5
A/Ontario/Rv3236/2016	54.0	99.2	101.7	218.0	10.0	51.7	43.8	10.0	19.2	10.0	36.9	41.1	10.0
A/Indiana/08/2011	34.4	72.5	110.4	177.3	22.4	43.1	46.8	31.4	10.0	40.6	36.4	10.0	10.0
A/Minnesota/11/2010	59.4	146.7	91.6	17.5	16.4	44.0	39.0	10.0	10.0	41.1	30.8	10.0	10.0
Sera	Sin16	Wis15	HK17	Nag17	Swe17	Mor17	Hel823	Hel941	Ohi17	Ohi12	Ont16	Ind11	Min11
A/Switzerland/9715293/2013	1291.6	939.6	1097.8	210.5	514.9	460.1	160.2	706.4	422.8	1072.5	354.5	227.5	302.5
A/Hong_Kong/4801/2014	636.5	780.7	1312.1	30.8	178.6	231.7	55.5	229.9	221.4	143.0	122.2	93.4	307.2
A/Texas/50/2012	436.8	556.3	462.7	17.2	242.5	288.5	53.2	656.4	73.0	295.0	482.4	16.0	149.6
A/Victoria/361/2011(Iv-165)	568.2	1881.1	1227.3	108.5	293.8	407.7	79.3	1203.0	290.1	561.0	793.8	125.4	393.1
A/Newcastle/67/2016	192.6	246.4	411.3	18.3	70.8	125.6	206.8	259.5	893.7	1553.0	1089.8	10.0	1619.5
A/Estonia/91621/2015	133.6	137.5	252.2	10.0	78.5	10.0	112.6	76.3	96.2	268.4	110.6	38.7	83.6
A/Utah/11/2011	212.4	753.0	215.9	41.4	138.5	265.5	100.4	549.1	208.9	1246.7	922.8	240.0	466.7
A/Gambia/G0071436/2012	390.8	532.3	694.8	37.3	174.6	139.2	296.9	470.4	233.7	154.4	423.5	157.8	709.3
A/Hanoi/Eli15597/2015	225.9	629.8	592.7	152.6	318.2	174.4	216.5	209.5	615.2	207.1	556.5	1205.8	669.2
A/Heilongjiang-Xiangyang/1134/2011	79.9	257.3	65.1	10.0	77.1	34.9	475.2	76.0	236.7	1330.2	70.6	48.1	127.2
A/Perth/16/2009nib-64	263.1	535.5	249.8	10.0	102.5	103.7	552.5	519.6	250.2	418.2	81.2	113.6	161.1
A/Tasmania/1018/2015	714.6	342.3	1149.8	178.8	321.5	551.2	363.1	1440.4	407.4	2702.2	819.1	463.8	689.9
A/Alaska/251/2015	655.6	1029.8	579.3	451.3	764.5	396.5	111.7	468.9	16.5	72.8	88.1	10.0	22.4
A/Singapore/Infimh-16-0019/2016	3746.2	1772.7	7210.9	1390.6	1637.5	1258.2	151.7	2123.3	94.6	282.6	68.0	10.0	122.3
A/Wisconsin/16/2015	772.0	5850.3	678.1	373.4	680.7	420.1	10.0	134.2	15.8	74.7	10.0	10.0	10.0
A/Hong_Kong/3089/2017	1534.5	708.7	4119.2	762.8	2114.3	744.7	297.3	324.9	10.0	10.0	10.0	10.0	10.0
A/Nagano/2153/2017	501.4	157.0	1131.7	960.8	258.6	313.2	94.3	115.9	85.6	136.8	10.0	10.0	54.6
A/Sweden/3/2017	5336.2	1167.5	7099.8	1132.6	8452.0	4595.1	506.4	2310.2	133.0	199.5	323.9	27.2	58.5
A/Moramanga/1907/2017	1060.6	273.0	3033.6	258.9	1286.0	6330.8	75.6	143.2	57.3	106.0	96.7	10.0	70.4
A/Helsinki/823/2013	519.3	215.9	745.1	10.0	326.8	113.1	5933.4	11949.0	33.1	87.9	85.5	32.4	46.8
A/Helsinki/941/2013	257.3	79.5	156.3	10.0	153.0	10.0	3729.4	17816.2	17.3	64.1	10.0	10.0	10.0
A/Ohio/13/2017	28.9	10.0	10.0	10.0	10.0	10.0	221.2	137.9	2567.9	28683.0	4819.3	781.1	4943.5
A/Ohio/62/2012	10.0	10.0	10.0	10.0	10.0	10.0	77.1	70.1	5592.0	3811.0	397.1	281.1	921.4
A/Ontario/Rv3236/2016	10.0	10.0	10.0	10.0	10.0	10.0	22.4	21.3	194.8	1500.5	6167.6	159.3	846.2
A/Indiana/08/2011	10.0	10.0	10.0	10.0	10.0	10.0	17.4	16.0	146.4	3741.8	486.8	2910.9	2498.3
A/Minnesota/11/2010	10.0	10.0	10.0	10.0	10.0	10.0	55.0	17.3	202.3	509.1	726.6	380.2	4375.6

Table S3.

NAI titers against H6N2 viruses of ferret sera obtained after the boost (Part I of II).

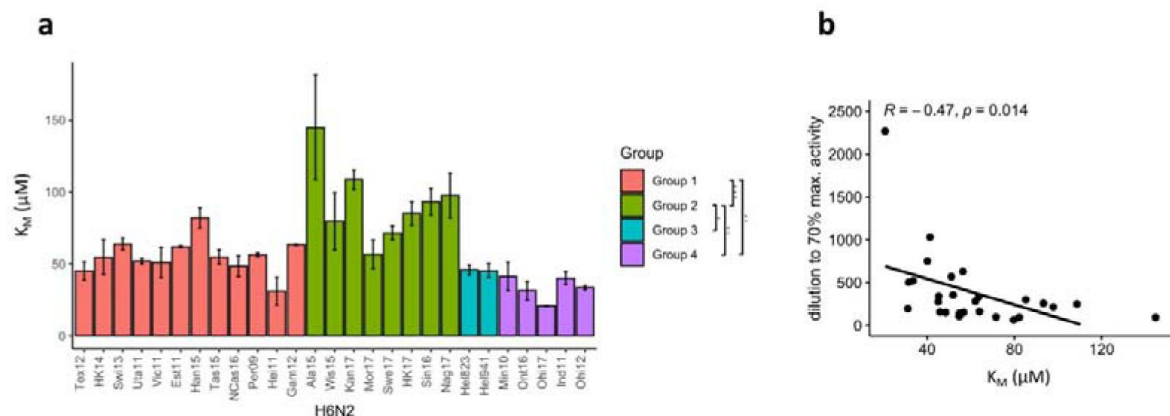


Figure 2. Figure supplement 1.

The N2 NA substrate affinity is distinct in the N2 phylogenetic groups.

(a) K_M was determined for each H6N2 reassortant virus using the fluorogenic substrate MUNANA. N2 NAs are coloured according to the phylogenetic groups. Global test comparing the 4 groups was done using ANOVA ($P = 2.4 \times 10^{-5}$) and pairwise comparison performed using Wilcoxon, $***P < 0.001$, $**P < 0.01$, $*P < 0.05$. (b) Scatter-plot and correlation between the K_M values and dilutions required to obtain 70% max activity in ELLA.

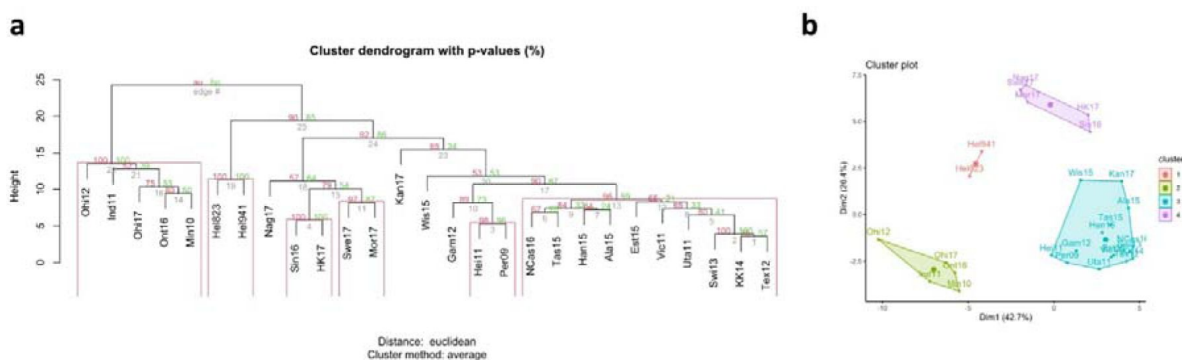


Figure 2. Figure supplement 2.

Antigenic relationship between NA of the H6N2 viruses.

(a) Hierarchical clustering based on multisampling and multiscale bootstrap resampling. Approximately unbiased p-values (AU) are shown in red, bootstrap values in green, and edge values in grey. The red brackets represent the clusters with AU > 95%. (b) K-means clustering represented using principal component analysis (k=4).

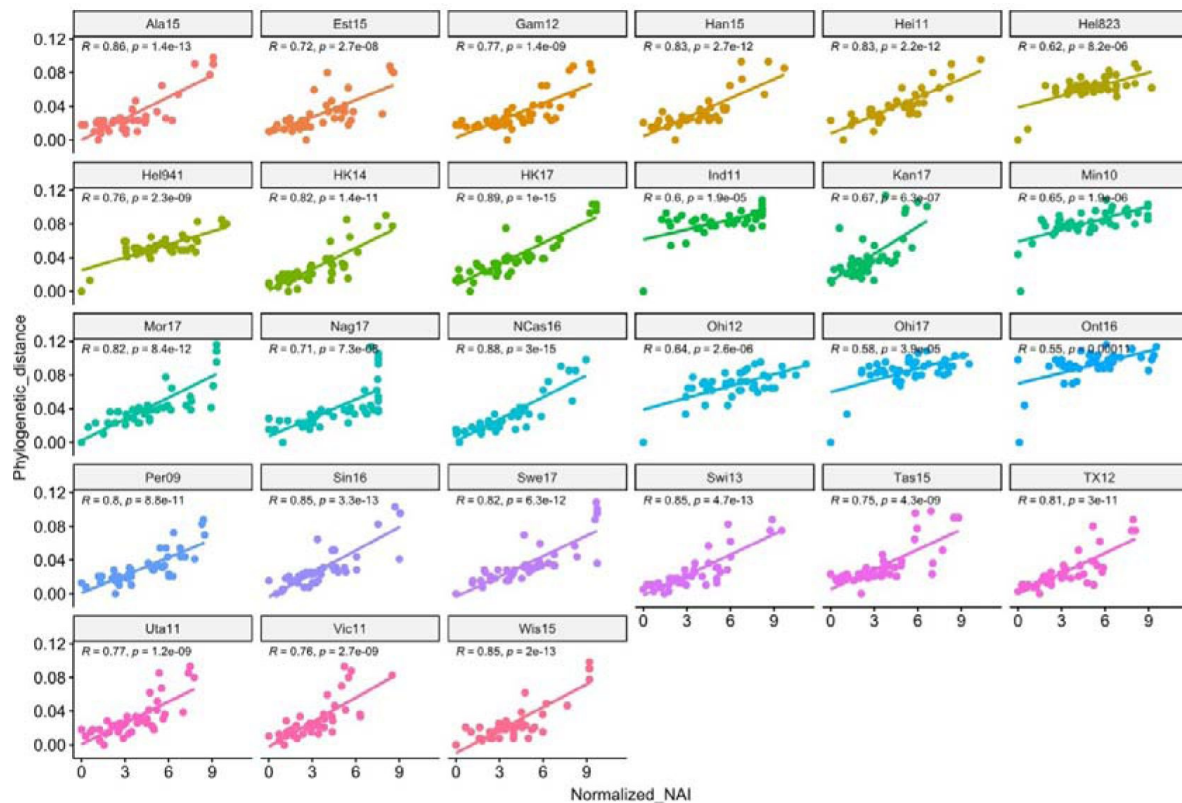


Figure 2. Figure supplement 3.

Correlation of NAI fold reduction and phylogenetic relatedness.

Scatter plot shows the phylogenetic distances (y-axis) and Normalized NAI as fold reduction relative to the maximal inhibition [$\log_2(\max \text{NAI}/\text{NAI})$] (x-axis). Each graph represents one of the 27 H6N2s that were tested.

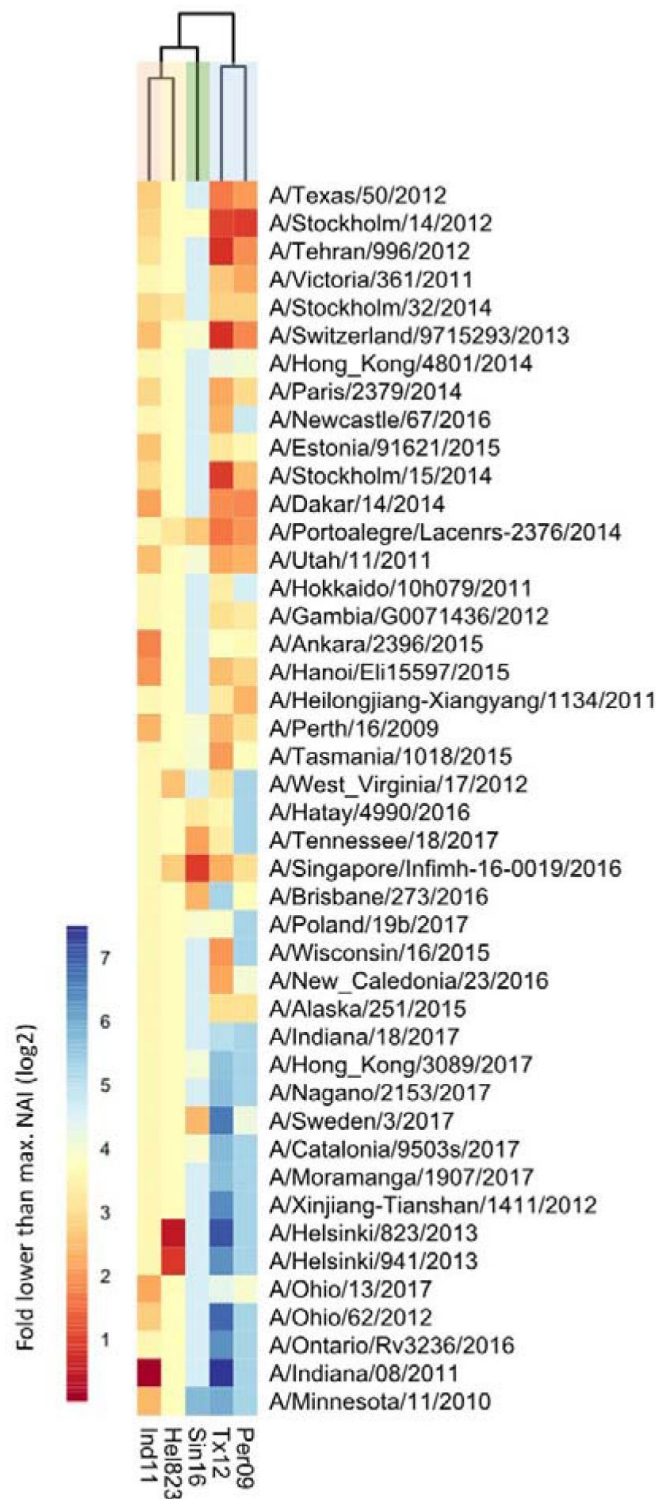


Figure 2. Figure supplement 4.

Breadth of N2 inhibition of sera from ferrets primed by H1N2 infection only.

Heat-map representing the average of normalized neuraminidase inhibition per H6N2 [log₂ (max NAI/NAI)] determined in ferret sera 2 weeks after infection. The red-to-blue scale indicates the fold reduction compared to the max NAI observed in ELLA against the H6N2 reassortants (listed at the bottom). UPGMA clustering of H6N2 inhibition profiles are shown on top of the heat map and coloured according to the phylogenetic groups.

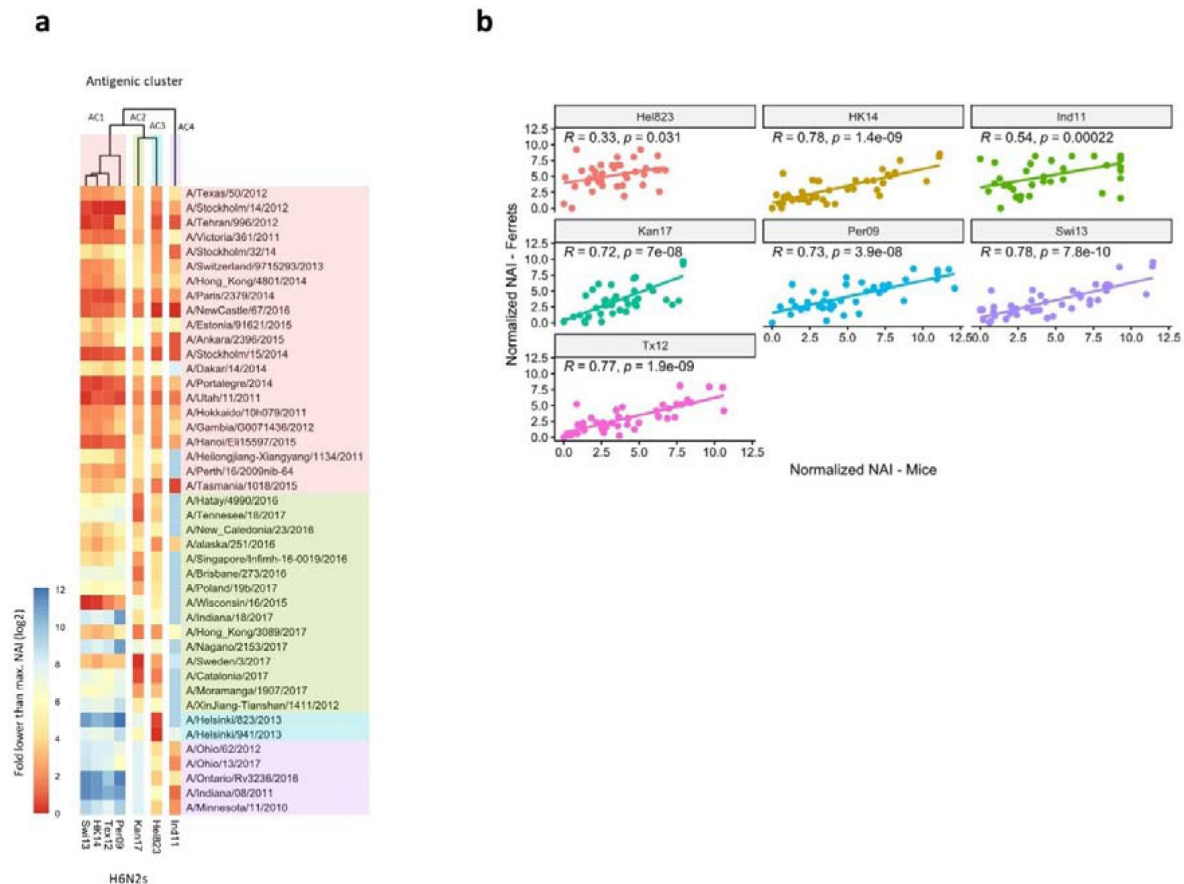


Figure 2. Figure supplement 5.

Breadth of N2 inhibition is confirmed in mouse sera.

Heat-map representing the average of normalized neuraminidase inhibition per H6N2 [$\log_2$ (max NAI/NAI)] determined in mice sera after the boost (listed on the right). The red-to-blue scale indicates low-to-high NAI observed in ELLA against the H6N2 reassortants (listed at the bottom). UPGMA clustering of H6N2 inhibition profiles are shown on top of the heat map and coloured according to the phylogenetic groups. (b) **Correlation of NAI between mice and ferrets.** Scatter plot shows the normalized NAI as fold reduction from maximal inhibition [$\log_2$ (max NAI/NAI)] from ferrets (y-axis) versus mice (x-axis). The 7 H6N2s tested are indicated in graph titers.

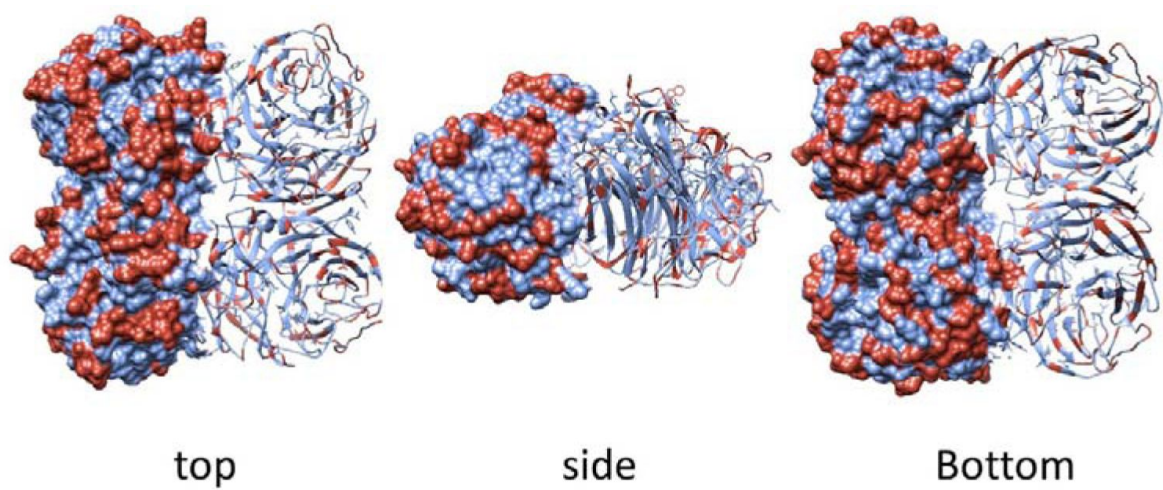


Figure 3 – figure supplement 2.

Geometrical distribution of amino acid substitutions in the N2 structure.

The surface and ribbon representation indicate the geometrical distribution of the amino acid substitutions that are present in the panel of 44 N2 NAs used as immunogens in the study. Substitutions are highlighted in red on the model structure of N2 from H2N2 (A/RI/5+/1957, PBD 4H53).

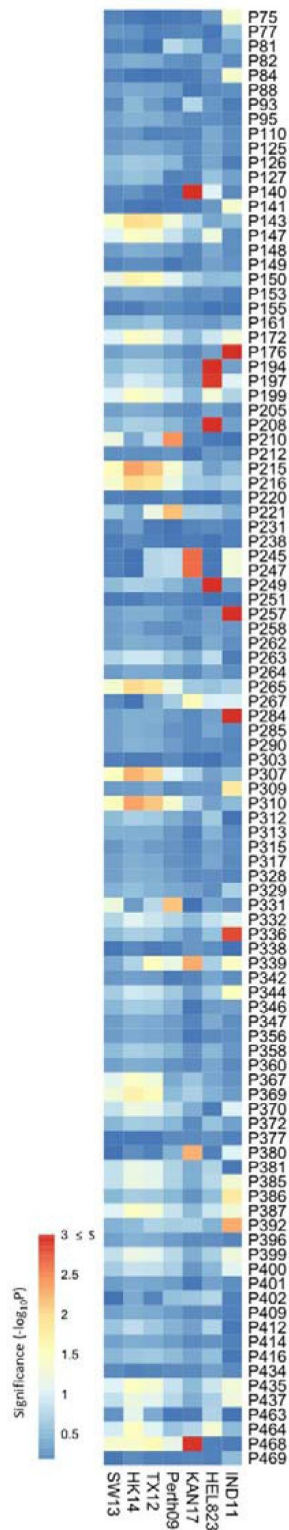


Figure 3 – figure supplement 3.

Association study was performed on data obtained from mouse sera.

(a) Association study performed for each H6N2 virus (listed at the bottom) to determine the probability of each variable amino acid (listed on the right) present in the panel of 44 NAs to impact NAI. The red-to-blue scale represents significance values ($-\log_{10}P$) associated with each variable amino acid.

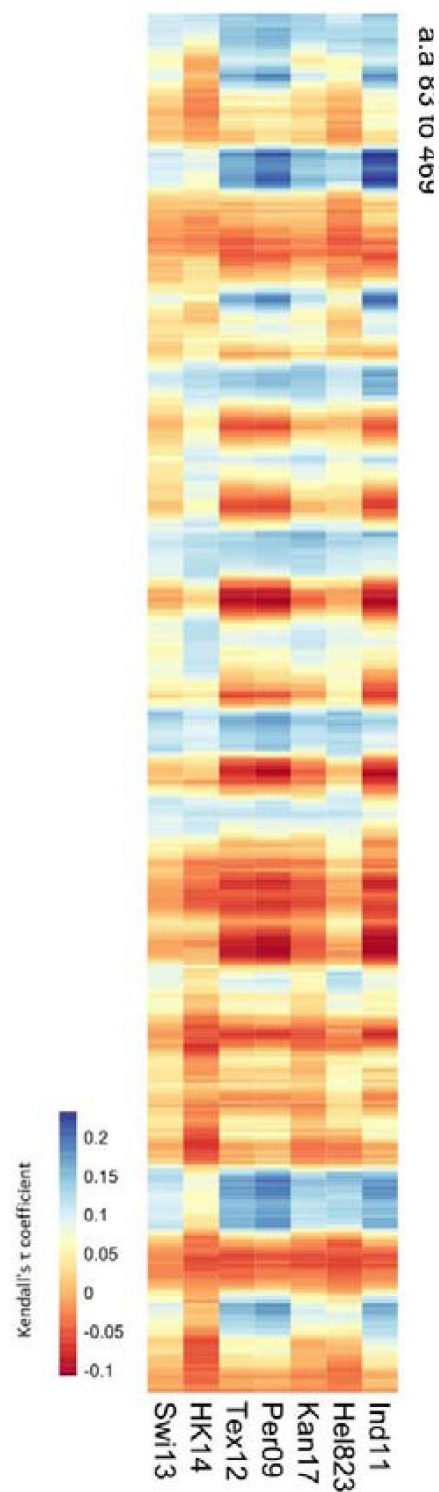


Figure 4 – figure supplement 1.

Correlation of values obtained from the mouse sera panel and amino acid distances.

(a) Significance values (102 variable amino acids) obtained from the association study were correlated to the distance between every amino acid and each of the variable amino acids in the NA structure. The red-to-blue scale represents the Tau values obtained in the Kendall correlation. Amino acids 83 to 469 are shown from top to bottom of the heatmap. The H6N2s used in the NAI panel are listed at the bottom.

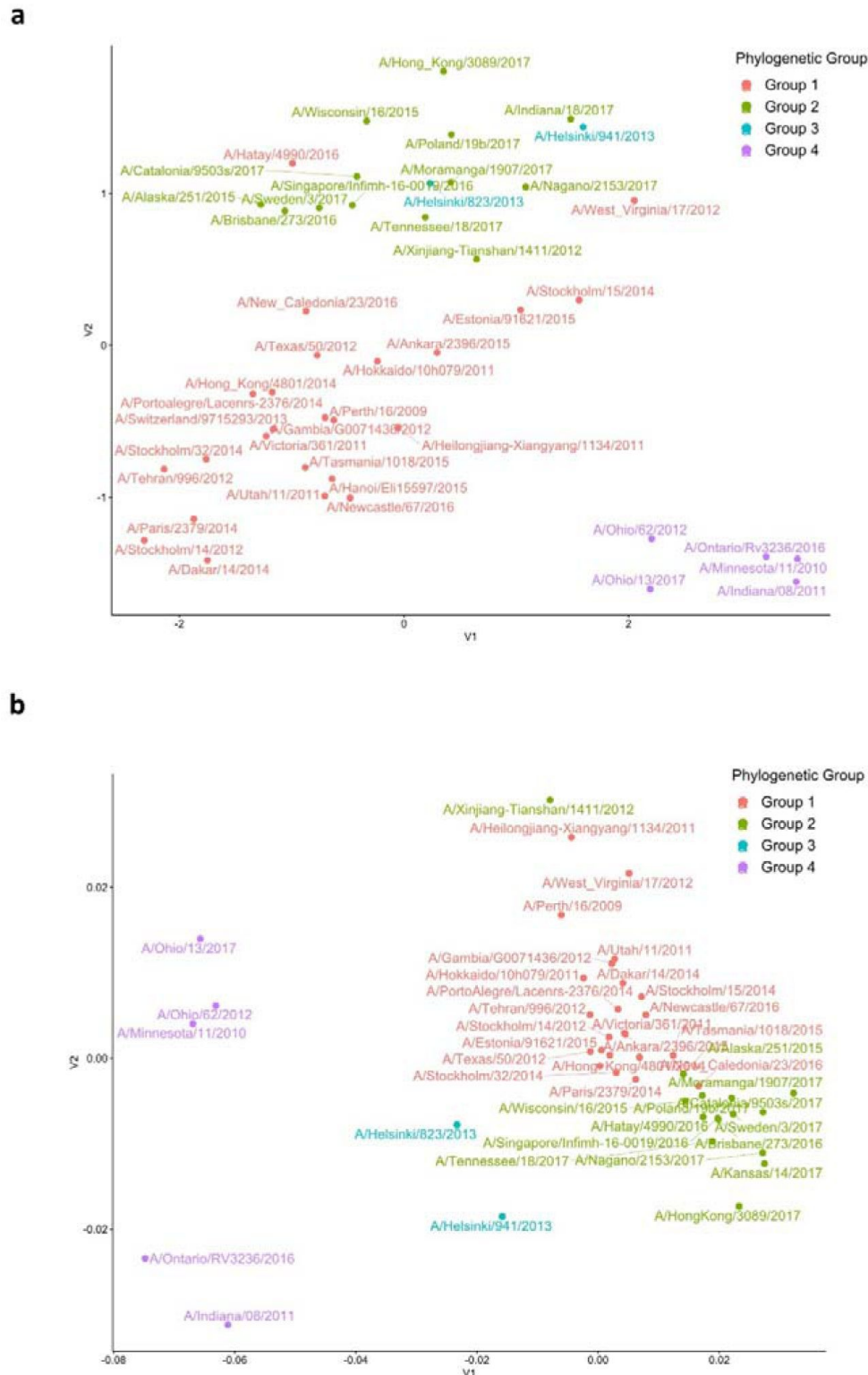


Figure 5 – figure supplement 1.

Multidimensional scaling defining antigenic and phylogenetic cartography of N2s.

(a) Average antigenic distances were plotted using classical MDS, N2 groups are colored according to the phylogenetic groups. (b) Pairwise comparison of amino acid sequences of the NA head domain was used to generate a matrix of phylogenetic differences that is represented using classical MDS. N2s are colored according to the phylogenetic groups.

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Editors

Reviewing Editor

Richard Neher

University of Basel, Switzerland

Reviewer #1 (Public Review):

Summary

The authors investigated the antigenic diversity of recent (2009- 2017) A/H3N2 influenza neuraminidases (NAs), the second major antigenic protein after haemagglutinin. They used 27 viruses and 43 ferret sera and performed NA inhibition. This work was supported by a subset of mouse sera. Clustering analysis determined 4 antigenic clusters, mostly in concordance with the genetic groupings. Association analysis was used to estimate important amino acid positions, which were shown to be more likely close to the catalytic site. Antigenic distances were calculated and a random forest model was used to determine potential important sites.

This has the potential to be a very interesting piece of work. At present, there are inconsistencies in the methods, results and presentation that limit its impact. In particular, there are weaknesses in some of the computational work.

Strengths

1. The data cover recent NA evolution and a substantial number (43) of ferret (and mouse) sera were generated and titrated against 27 viruses. This is laborious experimental work and is the largest publicly available neuraminidase inhibition dataset that I am aware of. As such, it will prove a useful resource for the influenza community.
2. A variety of computational methods were used to analyse the data, which give a rounded picture of the antigenic and genetic relationships and link between sequence, structure and phenotype.

Weaknesses

1. Inconsistency in experimental methods

Two ferret sera were boosted with H1N2, while recombinant NA protein for the others. This, and the underlying reason, are clearly explained in the manuscript. The authors note that boosting with live virus did not increase titres. Nevertheless, these results are included in the analysis when it would be better to exclude them (Figure 2 shows much lower titres to their own group than other sera).

2. Inconsistency in experimental results

Clustering of the NA inhibition results identifies three viruses which do not cluster with their phylogenetic group. Again this is clearly pointed out in the paper. Further investigation of this inconsistency is required to determine whether this has a genetic basis or is an experimental issue. It is difficult to trust the remaining data while this issue is unresolved.

3. Inconsistency in group labelling

A/Hatay/4990/2016 & A/New Caledonia/23/2016 are in phylogenetic group 1 in Figure 2 and phylogenetic group 1 in Figure 5 - figure supplement 1 panel a.

A/Kansas/14/2017 is selected as a representative of antigenic group 2, when in Figure 2 it is labelled as AC1 (although Figure 2 - supplement 4 which the text is referring to shows data for A/Singapore/Infimh-16-0019/2016 as the representative of AC2). A/Kansas/14/2017 is coloured and labelled as AC2 in Figure 2 - supplement 5.

The colouring is changed for Figure 3a at the bottom. A/Heilongjiang-Xiangyang/1134/2011 is coloured the same as AC4 viruses when it is AC1 in Figure 2.

This lack of consistency makes the figures misleading.

4. Data not presented, without explanation

The paper states that 44 sera and 27 H6N2 viruses were used (line 158). However, the results for the Kansas/14/2017 sera do not appear to be presented in any of the figures (e.g. Figure 2 phylogenetic tree, Figure 5 - figure supplement 1). It is not obvious why these data were not presented. The exclusion of this serum could affect the results as often the homologous titre is the highest and several heatmaps show the fold down from the highest titre.

5. The cMDS plot does not have sufficient quality assurance

A cMDS plot is shown in Figure 5 - figure supplement 1, generated using classical MDS. The following support for the appropriateness of this visualisation is not given.

- Goodness of fit of the cMDS projection, including per point and per titre.
- Testing of the appropriate number of dimensions (the two sera from phylogenetic group 3 are clustered with phylogenetic group 2; additional dimensions might separate these groups).
- A measure of uncertainty in positioning, e.g. bootstrapping.
- A sensitivity analysis of the assumption about titres below the level of detection (i.e. that $<20 = 10$).

Without this information, it is difficult to judge if the projection is reliable.

6. Choice of antigenic distance measure

The measure of antigenic distance used here is the average difference between titres for two sera. This is dependent on which viruses have been included in the analysis and will be biased by the unbalanced number of viruses in the different clusters (12, 8, 2, 5).

7. Association analysis does not account for correlations

For each H6N2 virus and position, significance was calculated by comparing the titres between sera that did or did not have a change at that position. This does not take into account the correlations between positions. For haemagglutinin, it can be impossible to determine the true antigenic effects of such correlated substitutions with mutagenesis studies.

8. Random forest method

25 features are used to classify 43 sera, which seems high ($p/3$ is typical for classification). By only considering mismatches, rather than the specific amino acid changes, some signals may be lost (for example, at a given position, one amino acid change might be neutral while another has a large antigenic effect). Features may be highly, or perfectly correlated, which will give them a lower reported importance and skew the results.

Reviewer #2 (Public Review):

Summary:

The authors characterized the antigenicity of N2 protein of 44 selected A(H3N2) influenza A viruses isolated from 2009-2017 using ferret and mice immune sera. Four antigenic groups were identified, which correlated with their respective phylogenetic/ genetic groups. Among 102 amino acids differed by the 44 selected N2 proteins, the authors identified residues that differentiate the antigenicity of the four groups and constructed a machine-learning model that provides antigenic distance estimation. Three recent A(H3N2) vaccine strains were tested in the model but there was no experimental data to confirm the model prediction results.

Strengths:

This study used N2 protein of 44 selected A(H3N2) influenza A viruses isolated from 2009-2017 and generated corresponding panels of ferret and mouse sera to react with the selected strains. The amount of experimental data for N2 antigenicity characterization is large enough for model building.

Weaknesses:

The main weakness is that the strategy of selecting 44 A(H3N2) viruses from 2009-2017 was not explained. It is not clear if they represent the overall genetic diversity of human A(H3N2) viruses circulating during this time. A comprehensive N2 phylogenetic tree of human A(H3N2) viruses from 2009-2017, with the selected 44 strains labeled in the tree, would be helpful to assess the representativeness of the strains included in the study. The second weakness is the use of double-immune ferret sera (post-infection plus immunization with recombinant NA protein) or mouse sera (immunized twice with recombinant NA protein) to characterize the antigenicity of the selected A(H3N2) viruses. Conventionally, NA antigenicity is characterized using ferret sera after a single infection. Repeated influenza exposure in ferrets has been shown to enhance antibody binding affinity and may affect the cross-reactivity to heterologous strains (PMID: 29672713). The increased cross-reactivity is supported by the NAI titers shown in Table S3, as many of the double immune ferret sera showed the highest reactivity not against its own homologous virus but to heterologous strains. Although the authors used the post-infection ferret sera to characterize 5 viruses (Figure 2, Figure Supplement 4), the patterns did not correlate well. If the authors repeat the NA antigenic analysis using the post-infection ferret sera with lower cross-reactivity, will the authors be able to identify more antigenic groups instead of 4 groups? Another weakness is that the authors used the newly constructed model to predict the antigenic distance of three recent A(H3N2) viruses but there is no experimental data to validate their prediction (eg. if these viruses are indeed antigenically deviating from group 2 strains as concluded by the authors).

Reviewer #3 (Public Review):**Summary:**

This paper by Portela Catani et al examines the antigenic relationships (measured using monotypic ferret and mouse sera) across a panel of N2 genes from the past 14 years, along with the underlying sequence differences and phylogenetic relationships. This is a highly significant topic given the recent increased appreciation of the importance of NA as a vaccine target, and the relative lack of information about NA antigenic evolution compared with what is known about HA. Thus, these data will be of interest to those studying the antigenic evolution of influenza viruses. The methods used are generally quite sound, though there are a few addressable concerns that limit the confidence with which conclusions can be drawn from the data/analyses.

Strengths:

- The significance of the work, and the (general) soundness of the methods.
- Explicit comparison of results obtained with mouse and ferret sera.

Weaknesses:

- Approach for assessing the influence of individual polymorphisms on antigenicity does not account for the potential effects of epistasis.
- Machine learning analyses were neither experimentally validated nor shown to be better than simple, phylogenetic-based inference.